

# 재생에너지 기술 및 응용

(Renewable Energy Technologies and Applications)

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**H**ydrogen **E**nerg**Y** Lab **ON** **E**arth

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# Course Introduction

## ❑ Lecture: Renewable Energy Technologies and Applications (L01211101)

– Class time: Tuesday, 12:00–14:50

## ❑ Professor: Haewon Seo

– Office: Prof Res Bldg , Rm 202

– E-mail: heyone@hufs.ac.kr

– Office hour: Wednesday (till 17:00)



## ❑ Scope

– This course explores the necessity of renewable energy in meeting the surging power demands of the AI era. Grounded in thermodynamics and fluid mechanics, it examines the operating principles and technologies of major renewable systems. Furthermore, it reviews the latest technological trends and discusses system integration strategies for practical applications.

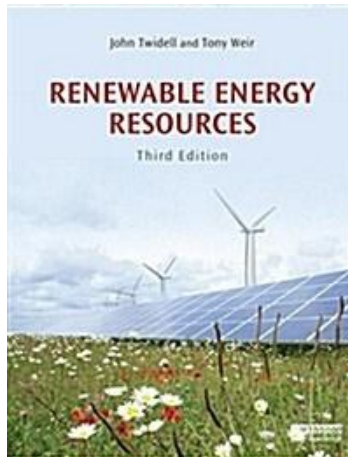
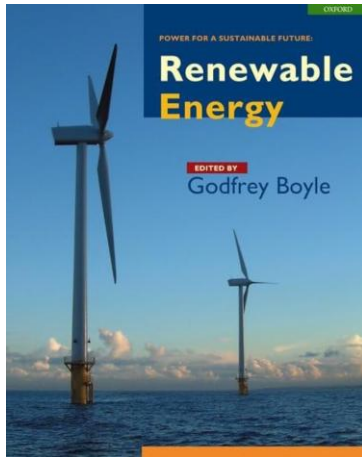
# Course Introduction



## □ Textbook

### – Main

- Renewable Energy (G. Boyle), 2<sup>nd</sup> Ed., Oxford University Press (2004)
- Renewable Energy Sources, (J. Twidell & T. Weir), 3<sup>rd</sup> Ed., Talyor & Francis (2015)
- Thermodynamics (Y.A. Cengel et al.), 9<sup>th</sup> Ed., McGraw-Hill (2018)



### – Reference

- Scientific articles
- IPCC, Corpernicus Climate Change, NOAA, NASA, etc.

# Course Introduction

□ **Schedule;** to be flexibly operated

| Week                   | Date                            | Contents                         |
|------------------------|---------------------------------|----------------------------------|
| <b>01<sup>st</sup></b> | <b>Sep 01<sup>st</sup> 2026</b> | <b>Introduction to Energy</b>    |
| 02 <sup>nd</sup>       | Sep 08 <sup>th</sup> 2026       | Fundamentals of Thermodynamics   |
| 03 <sup>rd</sup>       | Sep 15 <sup>th</sup> 2026       | Fossil Fuels                     |
| 04 <sup>th</sup>       | Sep 22 <sup>nd</sup> 2026       | Solar Thermal Energy             |
| 05 <sup>th</sup>       | Sep 29 <sup>th</sup> 2026       | Solar Photovoltaics (1)          |
| 06 <sup>th</sup>       | Oct 06 <sup>th</sup> 2026       | Solar Photovoltaics (2)          |
| 07 <sup>th</sup>       | Oct 13 <sup>th</sup> 2026       | Presentation (1)                 |
| 08 <sup>th</sup>       | Oct 20 <sup>th</sup> 2026       | Midterm Exam                     |
| 09 <sup>th</sup>       | Oct 27 <sup>th</sup> 2026       | Fundamentals of Fluid Dynamics   |
| 10 <sup>th</sup>       | Nov 03 <sup>rd</sup> 2026       | Wind Energy                      |
| 11 <sup>th</sup>       | Nov 10 <sup>th</sup> 2026       | Hydropower                       |
| 12 <sup>th</sup>       | Nov 17 <sup>th</sup> 2026       | Bioenergy                        |
| 13 <sup>th</sup>       | Nov 24 <sup>th</sup> 2026       | Waste Energy & Geothermal Energy |
| 14 <sup>th</sup>       | Dec 01 <sup>st</sup> 2026       | Presentation (2)                 |
| 15 <sup>th</sup>       | Dec 15 <sup>th</sup> 2026       | Final Exam                       |

# Course Introduction

## Evaluation

- Attendance (10%)
- Assignment (20%)
- Midterm Exam (30%)
- Final Exam (40%)

# 1. Introduction to Energy

# What is Energy?



(a) Refrigerator



(b) Boats



(c) Aircraft and spacecraft



(d) Power plants



(e) Human body



(f) Cars



(g) Wind turbines



(h) Food processing



(i) A piping network in an industrial facility.

# What is Energy?

## Energy

The quantitative property transferred to a system or body, recognizable as the capacity to do work, produce change, or emit heat and light

- Energy is a scalar quantity (has magnitude, no direction)
- Total energy in a closed system is conserved over time

$$\frac{dE_{\text{total}}}{dt} = 0 \Leftrightarrow E_{\text{total}} = \text{constant}$$

- Units: J (SI), Wh, cal, eV, etc.

1 Wh = 3600 J (Electrical energy)

1 cal = 4.184 J (Metabolic energy)

1 eV =  $1.602 \times 10^{-19}$  J (Particle/Subatomic energy)

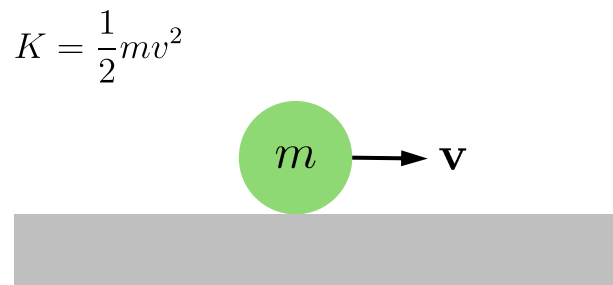
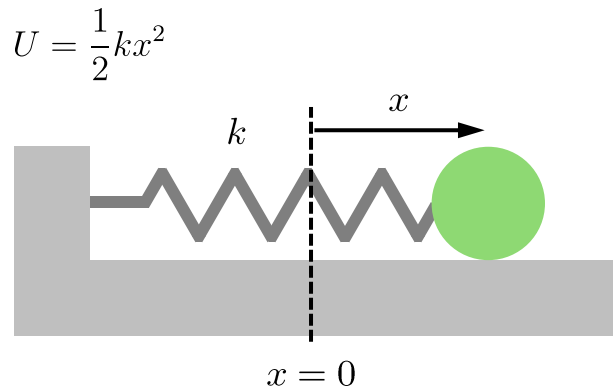


# Forms of Energy

## Macroscopic Forms

Possessed by a system as a whole relative to an external reference frame

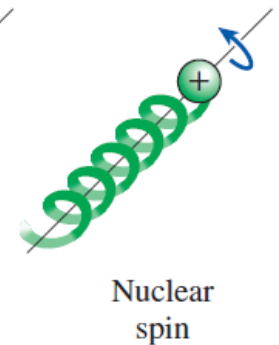
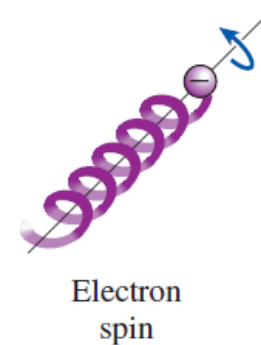
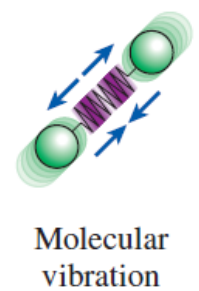
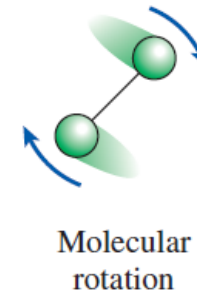
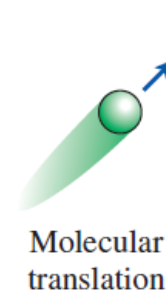
(e.g.) Potential energy, Kinetic Energy



## Microscopic Forms

Related to the molecular structure and activity of a system, and independent of external reference frames.

The sum of all the microscopic forms of energy is called *the internal energy* of a system.



# Forms of Energy

## ☐ Potential Energy

Stored energy that depends on the relative positions of objects

– Chemical Energy  
(e.g., Gasoline, Food)



– Mechanical Energy  
(e.g., Compressed spring)



– Nuclear Energy  
(e.g., Uranium fuel)



– Gravitational Energy  
(e.g., Water behind a dam)



## ☐ Kinetic Energy

Energy that a system possesses as a result of its motion relative to some reference frame

– Radiant Energy  
(e.g., Sunlight, Lamp light)



– Thermal Energy  
(e.g., Boiling water)



– Motion Energy  
(e.g., A moving car)



– Sound Energy  
(e.g., A clapping sound)



# International System of Units

## SI units

A coherent system of units of measurement starting with seven base units, including **kg** (mass), **m** (length), **s** (time), **A** (electric current), **K** (temperature), **cd** (luminous intensity), and **mol** (amount of matter)

### MLT dimension system

- $M \Leftrightarrow$  kilogram (kg),  $L \Leftrightarrow$  meter (m),  $T \Leftrightarrow$  second (s)
- (e.g.) [Velocity] =  $L T^{-1} \Leftrightarrow m s^{-1}$ , [Volumetric flow] =  $L^3 T^{-1} \Leftrightarrow m^3 s^{-1}$

| Variables     | Unit (SI)          | MLT                      |
|---------------|--------------------|--------------------------|
| Acceleration  | $m s^{-2}$         | $LT^{-2}$                |
| Angle         | rad                | $M^0 L^0 T^0$            |
| Area          | $m^2$              | $L^2$                    |
| Density       | $kg m^{-3}$        | $ML^{-3}$                |
| Energy        | J                  | $ML^2 T^{-2}$            |
| Force         | N                  | $MLT^{-2}$               |
| Frequency     | Hz                 | $T^{-1}$                 |
| Heat capacity | $J kg^{-1} K^{-1}$ | $L^2 T^{-2} \Theta^{-1}$ |
| Length        | m                  | $L$                      |
| Mass          | kg                 | $M$                      |

| Variables        | Unit (SI)    | MLT              |
|------------------|--------------|------------------|
| Power            | W            | $ML^2 T^{-3}$    |
| Pressure         | Pa           | $ML^{-1} T^{-2}$ |
| Strain           | –            | $M^0 L^0 T^0$    |
| Stress           | Pa           | $ML^{-1} T^{-2}$ |
| Surface tension  | $N m^{-1}$   | $MT^{-2}$        |
| Temperature      | K            | $\Theta$         |
| Time             | s            | $T$              |
| Viscosity (dyn.) | $Pa s$       | $ML^{-1} T^{-1}$ |
| Viscosity (kin.) | $m^2 s^{-1}$ | $L^2 T^{-1}$     |
| Volume           | $m^3$        | $L^3$            |

# International System of Units

## □ Metric Prefix

A unit prefix that precedes a basic unit of measure to indicate a multiple or submultiple of the unit

| Multiple  | Prefix    | Adoption |
|-----------|-----------|----------|
| $10^{30}$ | quetta, Q | 2022     |
| $10^{27}$ | ronna, R  |          |
| $10^{24}$ | yotta, Y  | 1991     |
| $10^{21}$ | zetta, Z  |          |
| $10^{18}$ | exa, E    | 1975     |
| $10^{15}$ | peta, P   |          |
| $10^{12}$ | tera, T   | 1960     |
| $10^9$    | giga, G   |          |
| $10^6$    | mega, M   | 1873     |
| $10^3$    | kilo, k   | 1795     |
| $10^2$    | hecto, h  |          |
| $10^1$    | deka, da  |          |

| Multiple   | Prefix       | Adoption |
|------------|--------------|----------|
| $10^{-1}$  | deci, d      | 1795     |
| $10^{-2}$  | centi, c     |          |
| $10^{-3}$  | mili, m      |          |
| $10^{-6}$  | micro, $\mu$ | 1873     |
| $10^{-9}$  | nano, n      | 1960     |
| $10^{-12}$ | pico, p      |          |
| $10^{-15}$ | femto, f     | 1964     |
| $10^{-18}$ | atto, a      |          |
| $10^{-21}$ | zepto, z     | 1991     |
| $10^{-24}$ | yocto, y     |          |
| $10^{-27}$ | ronto, r     | 2022     |
| $10^{-30}$ | quecto, q    |          |

# Force and Energy

## □ Force

Any action that tends to maintain or alter the motion of a body or to distort it

$$\mathbf{F} = \frac{d\mathbf{p}}{dt} = m\mathbf{a} \quad (\text{Newton's 2}^{\text{nd}} \text{ Law})$$

- **[Force] = [Mass][Acceleration] =  $\text{MLT}^{-2} \Leftrightarrow \text{kg m s}^{-2} = \text{N}$**

e.g., the gravitational forces on an 80.0-kg man on Earth and the Moon are respectively

$$F_{\text{g,Earth}} = 80.0 \text{ kg} \times 9.81 \text{ m s}^{-2} = 785 \text{ N} = 80.0 \text{ kgf}$$

$$F_{\text{g,Moon}} = 80.0 \text{ kg} \times 1.62 \text{ m s}^{-2} = 130 \text{ N} = 13.1 \text{ kgf}$$

## □ Energy

The capacity for performing work

$$W = \int \mathbf{F} \cdot d\mathbf{r}$$

- **[Energy] = [Force][Length] =  $\text{ML}^2\text{T}^{-2} \Leftrightarrow \text{kg m}^2 \text{s}^{-2} = \text{J}$**

e.g., the kinetic energy of an 80.0-kg man who runs at  $10.0 \text{ m s}^{-1}$  is

$$K = 1/2 \times 80.0 \text{ kg} \times (10.0 \text{ m s}^{-1})^2 = 4000 \text{ J} \quad (\text{c.f., daily adult caloric intake} = \sim 10 \text{ MJ})$$

# Power and Pressure

## Power

The rate at which work is performed

$$P = \frac{dW}{dt}$$

- [Power] = [Energy][Time]<sup>-1</sup> = ML<sup>2</sup>T<sup>-3</sup> ⇔ J s<sup>-1</sup> = W

e.g., the power an 80.0-kg man exerts climbing stairs at 1.00 m s<sup>-1</sup> is

$$P = 80.0 \text{ kg} \times 9.81 \text{ m s}^{-2} \times 1.00 \text{ m s}^{-1} = 785 \text{ W} \quad (\text{c.f., } 1 \text{ horsepower} = 746 \text{ W})$$

## Pressure

The perpendicular force per unit area, or the stress at a point within a confined fluid

$$p = \frac{F}{A}$$

- [Pressure] = [Force][Area]<sup>-2</sup> = ML<sup>-1</sup>T<sup>-2</sup> ⇔ N m<sup>-2</sup> = Pa

e.g., the pressure exerted on the ground by an 80.0-kg man with a total contact area of 0.0450 m<sup>2</sup> is

$$p = 80.0 \text{ kg} \times 9.81 \text{ m s}^{-2} / 0.0450 \text{ m}^2 = 17.4 \text{ kPa} \quad (\text{c.f., } p_{\text{atm}} = 101.325 \text{ kPa})$$

# Power Generation



## ❑ Quiz (1)

A household is paying KRW 215  $\text{kWh}^{-1}$  for electric power. To reduce its power bill, the household installed a solar panel with a rated power of 3 kW. If the solar panel operates 1400 hours per year ( $\sim 4$  hours per day) at the rated power, determine *the amount of electric power generated by the solar panel and the money saved by the household per year*.

$$\begin{aligned}\text{Total energy} &= \text{Energy per unit time} \times \text{Time interval} \\ &= \\ &= \quad \text{kWh yr}^{-1}\end{aligned}$$

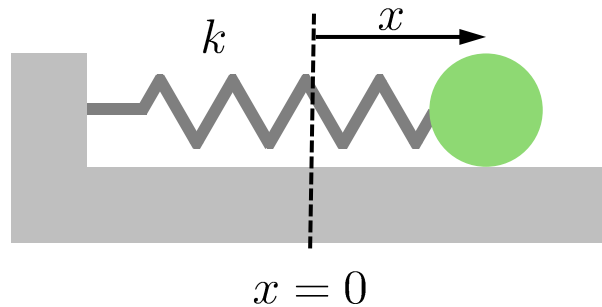
$$\begin{aligned}\text{Money saved} &= \text{Total energy} \times \text{Unit cost of energy} \\ &= \\ &= \text{KRW} \quad \text{yr}^{-1}\end{aligned}$$



# Energy Conservation

## Quiz (2)

Consider a one-dimensional motion along the  $x$ -axis of a spring–mass system, where a ball of mass  $m$  is attached to a spring with a spring constant  $k$ . Assume that no nonconservative forces are acting on the system. \*friction, air resistance



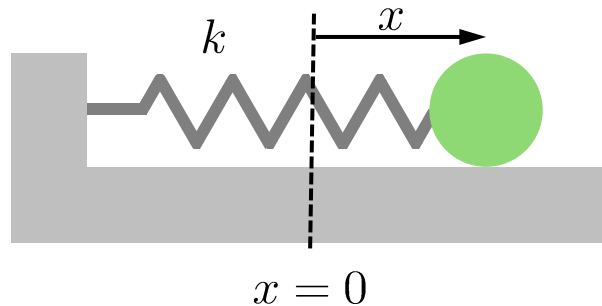
- Write down *the equation of motion* for the system.
- Determine *the angular frequency  $\omega$  of the oscillation* using  $k$  and  $m$ .



# Energy Conservation

## Quiz (2) (Cont'd)

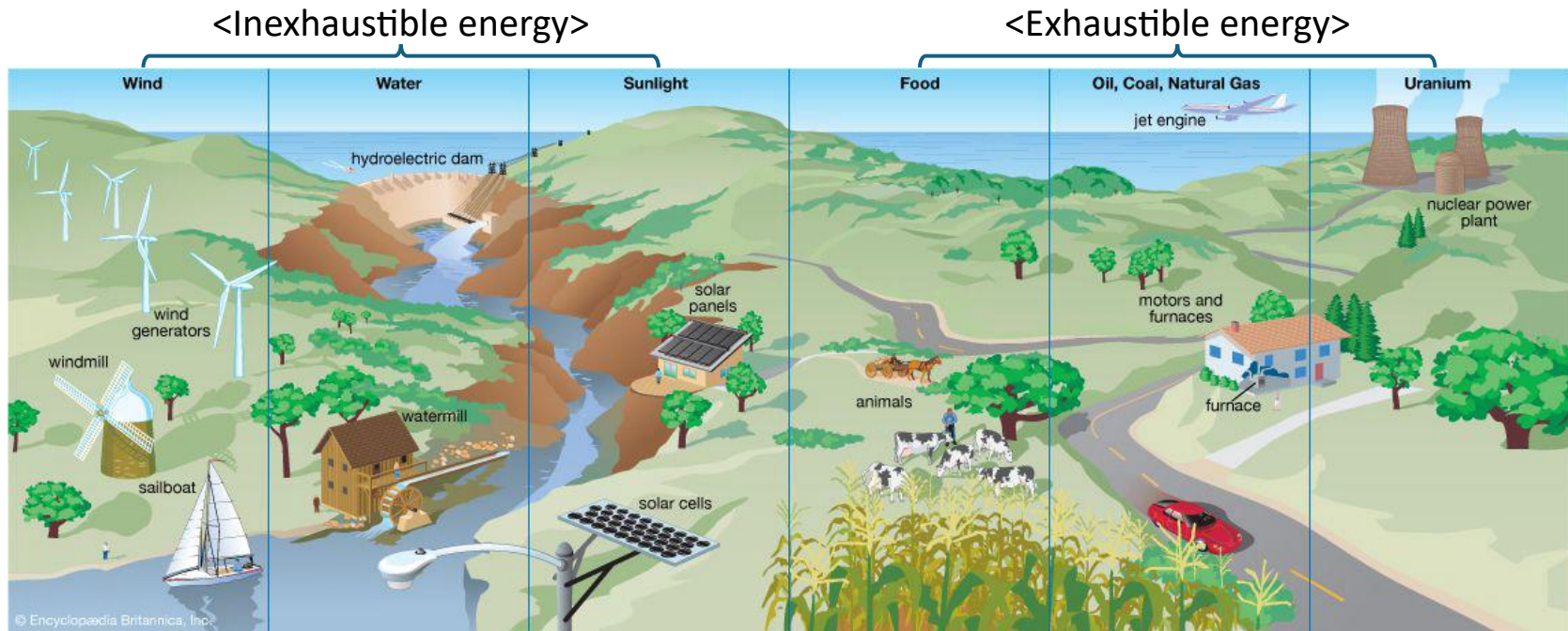
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- Derive *the displacement*  $x(t)$  using the amplitude  $x_0$ ,  $k$ , and  $m$ .
- Estimate *the total energy of the system*  $E_{\text{total}}$  and prove that it is constant over time.

# Energy in Daily

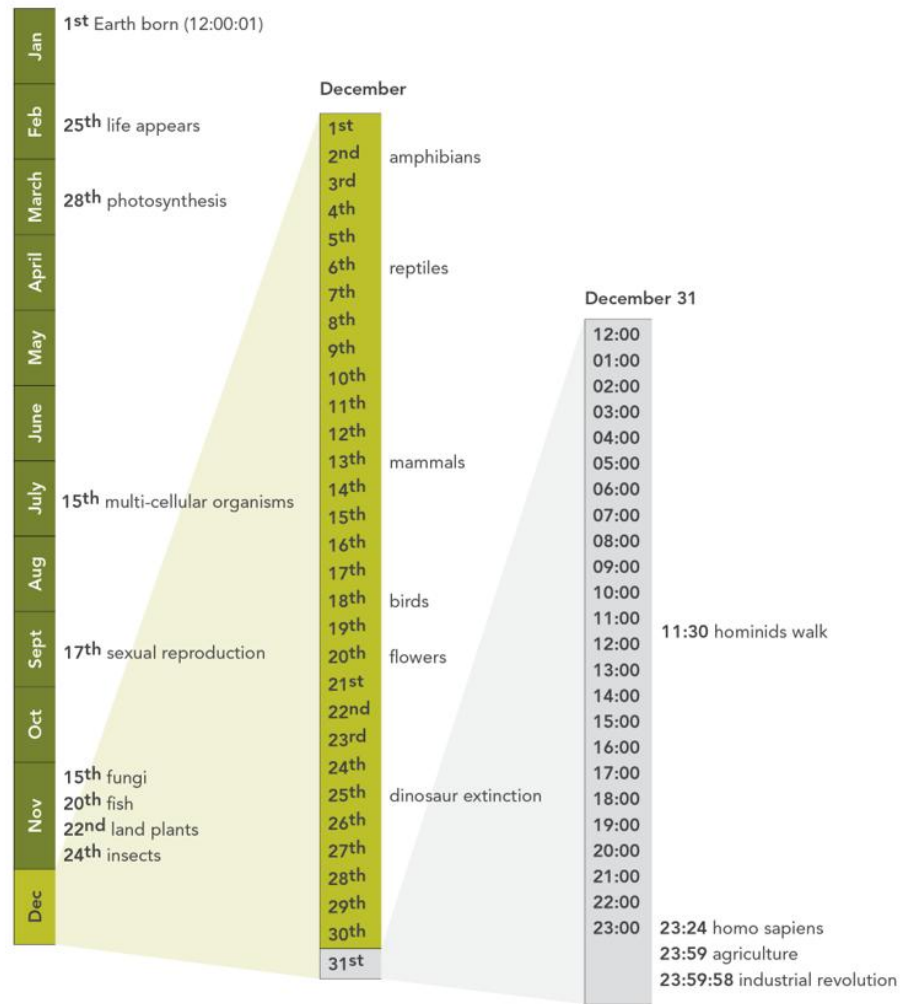
## ☐ Energy Resources



- Harnessing the power of *nature* (e.g., wind, water, and sunlight)
- Foods provided by *plants and animals*
- Burning *fossil fuels* (e.g., oil, coal, and natural gas) for energy
- Nuclear energy from *radioactive atoms* (e.g., uranium, thorium, etc.)

# Timeline of Energy

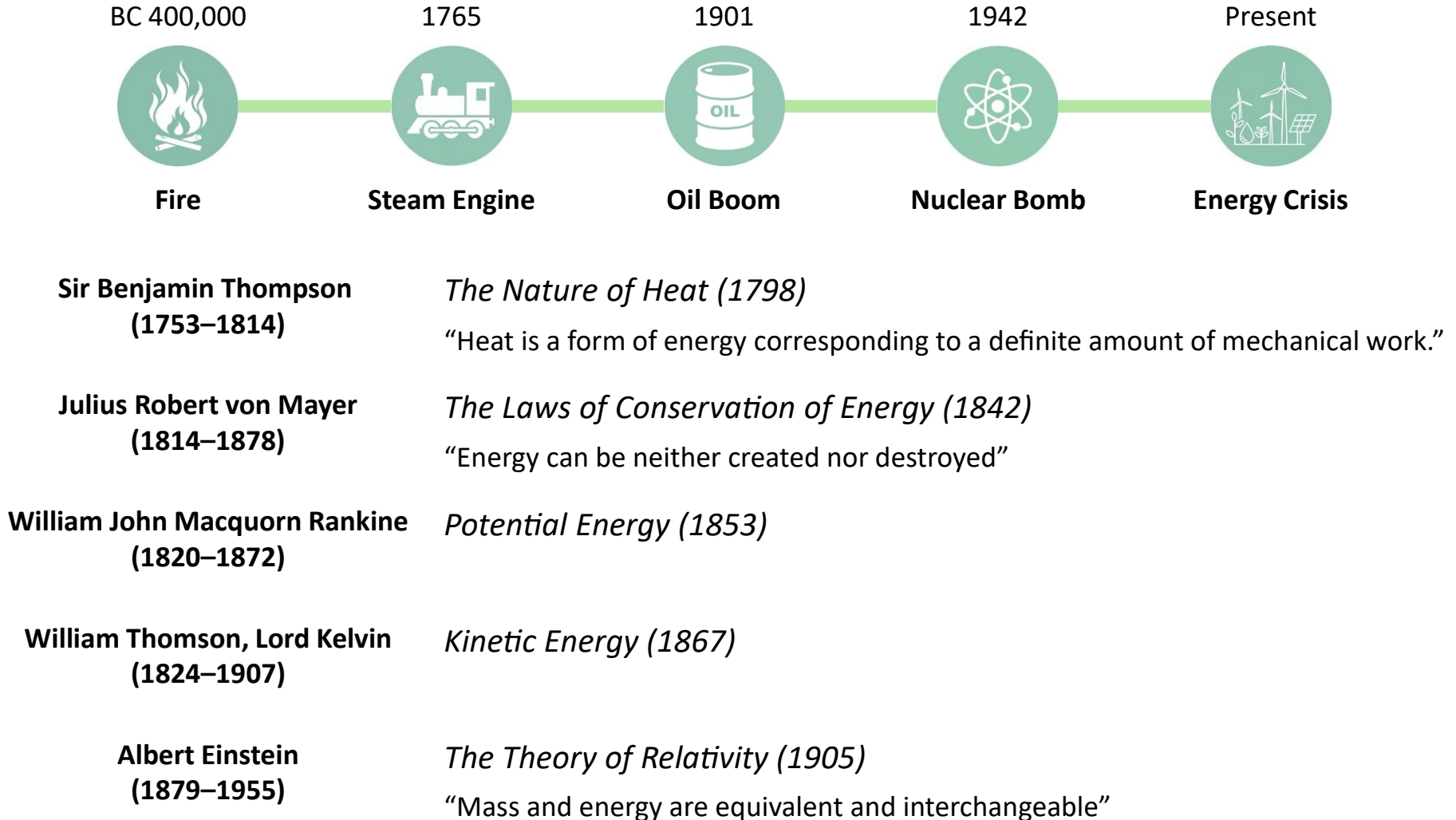
## Geologic Calendar



<Source: Biomimicry 3.8>

# Timeline of Energy

## □ Energy Transition



# Types of Energy

## ❑ Nonrenewable Energy

Energy derived from natural resources that cannot be readily replenished in a short period

*\*Millions of years to be formed*



Oil



Coal



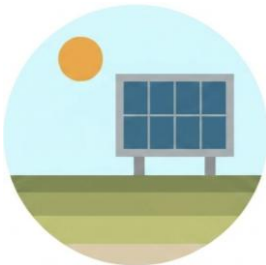
Natural Gas



Nuclear

## ❑ Renewable Energy

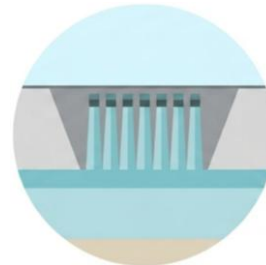
Energy derived from natural resources that are replenished on a human timescale



Solar



Wind



Hydropower



Biomass

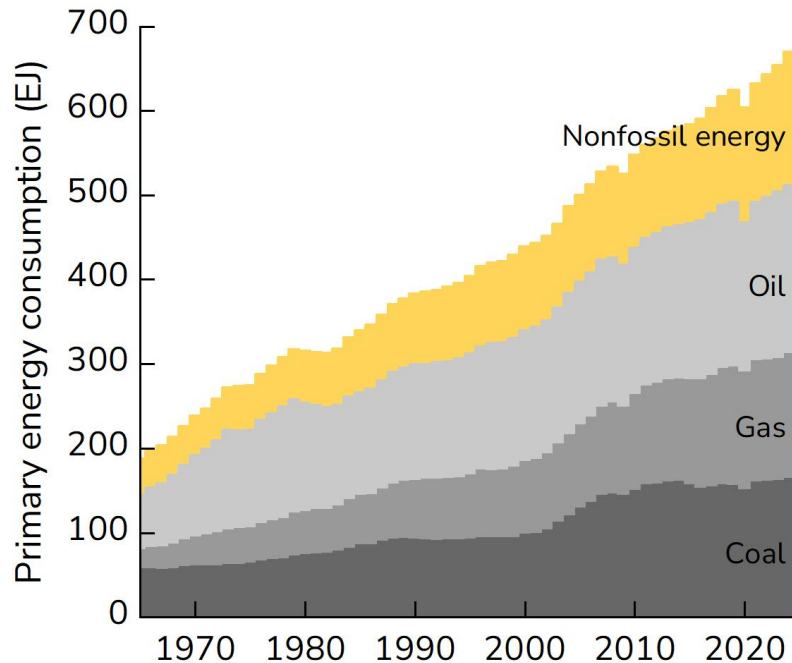


Geothermal Heat

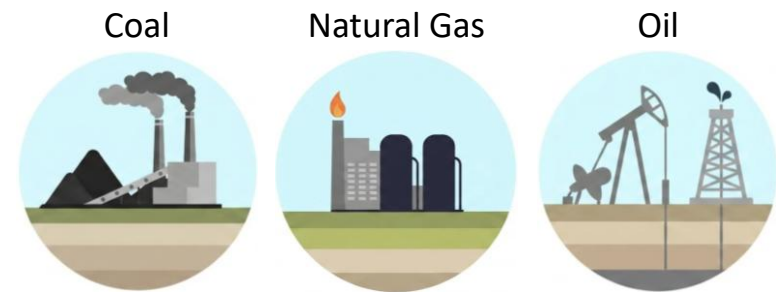
# Types of Energy

## Global Energy Consumption

The world relies on fossil fuels, which account for ~80% of primary energy consumption



<Source: Statistical Review of World Energy (2025)>



|                     | Coal                          | Natural Gas                   | Oil                           |
|---------------------|-------------------------------|-------------------------------|-------------------------------|
| Energy Use (2024)   | 165 EJ (~25%)                 | 149 EJ (~22%)                 | 199 EJ (~30%)                 |
| Left (2020)         | 139 yr                        | 49 yr                         | 56 yr                         |
| Efficiency (2024)   | 33%                           | 44%                           | 31%                           |
| GHG emission (2024) | 15.8×10 <sup>9</sup> t (~41%) | 8.01×10 <sup>9</sup> t (~21%) | 12.5×10 <sup>9</sup> t (~32%) |

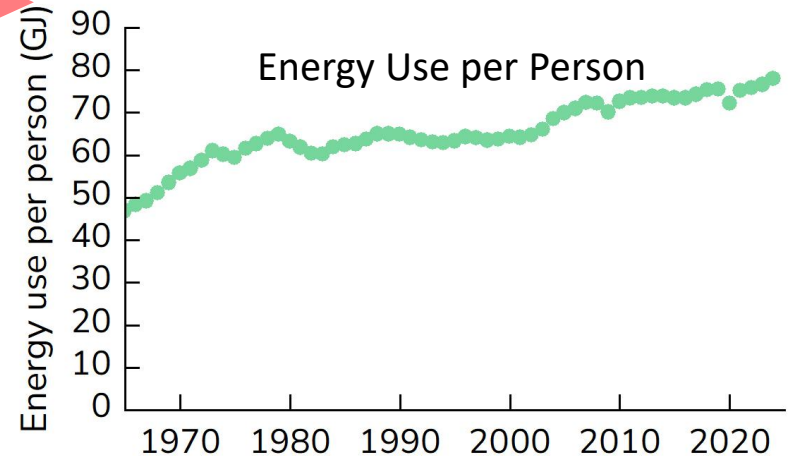
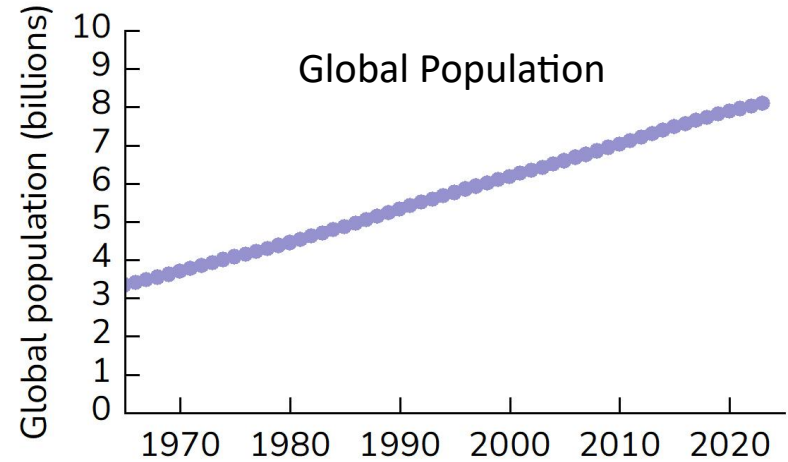
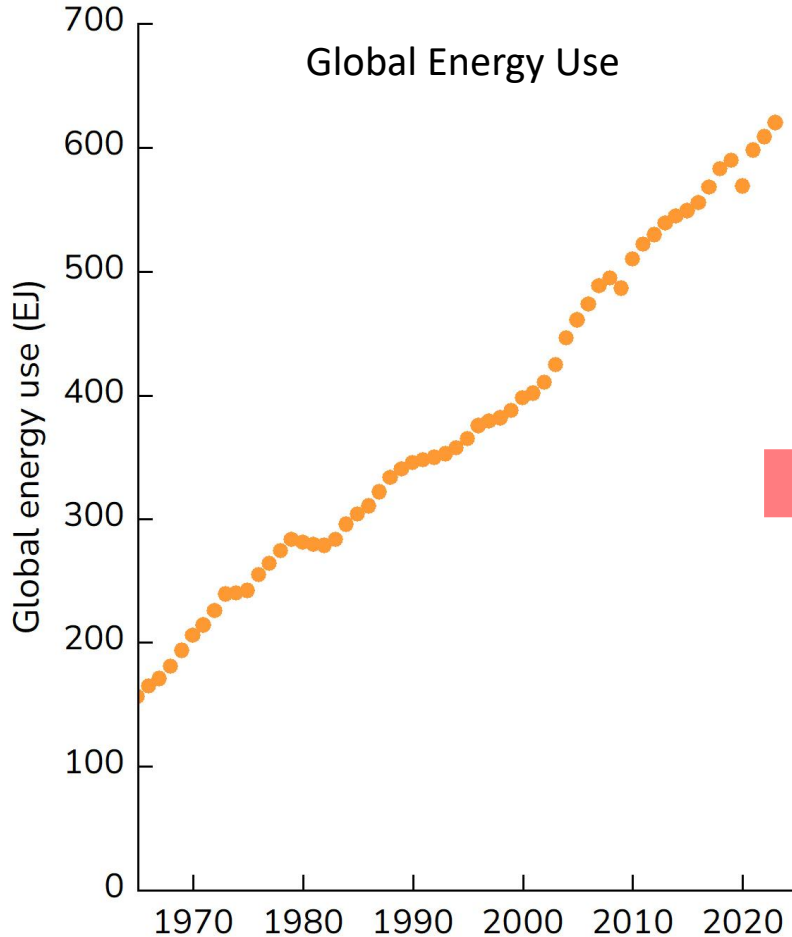
<Source: Form EIA-923 (2026), Global Carbon Budget (2025)>

- Political issues: Foreign sources, Resource wars, Carbon taxes
- Economical issues: Increased global demand, Limited resource
- Environmental issues: Climate crisis due to CO<sub>2</sub>, NO<sub>x</sub>, smog, etc.

# Types of Energy



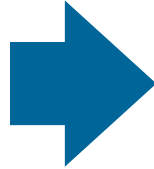
## Global Energy Use



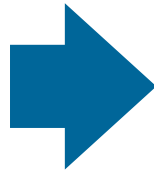
– More and more energy will be needed in near future



# Types of Energy



Transformation



Transformation



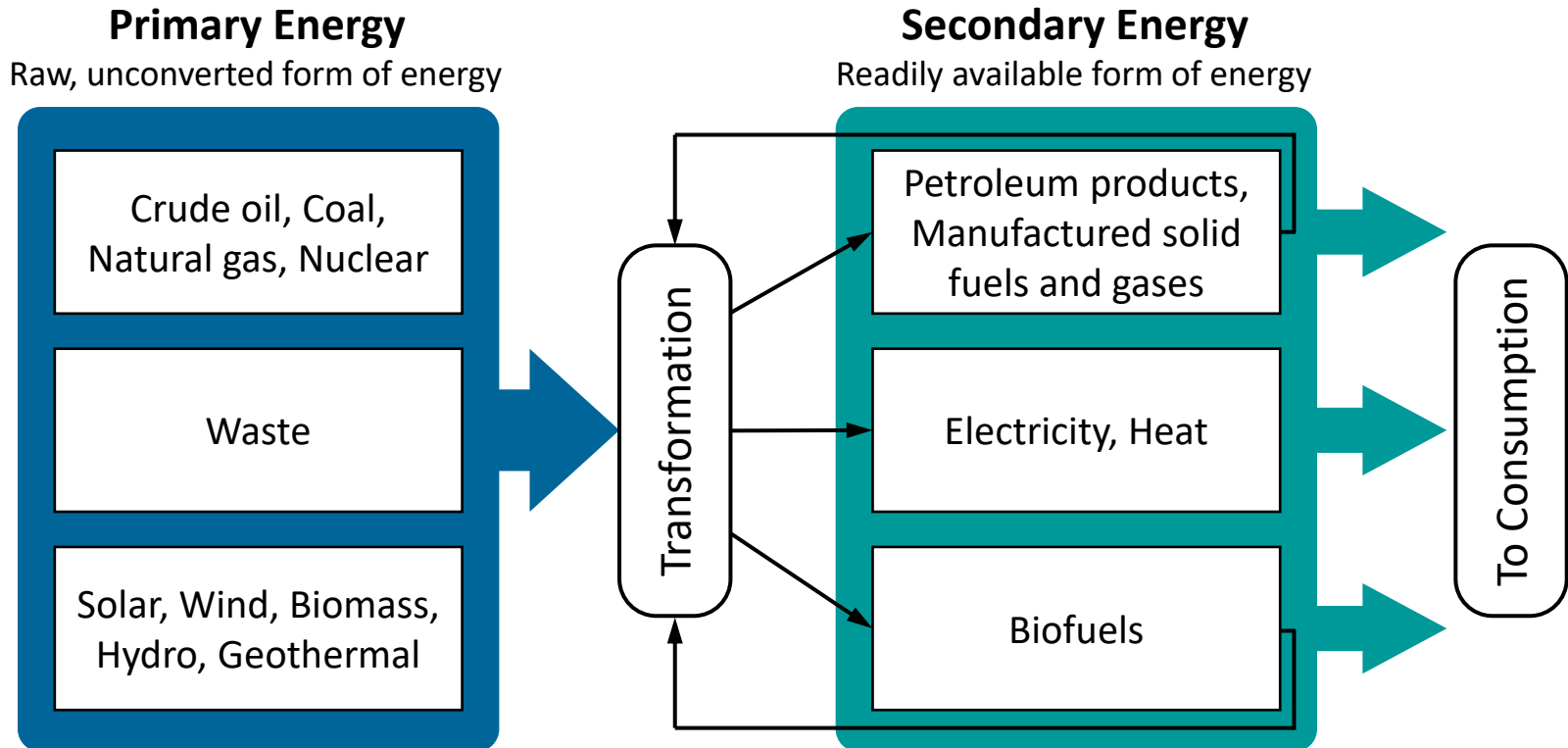
– Transforming a raw, unconverted form into a useful form is necessary



# Types of Energy

## □ Primary and Secondary Energy

Energy is indeed abundant, but *usable* energy is strictly limited, making *tools* essential



- *Usable* energy must be transformed from primary energy sources
- Every energy conversion process must entail *losses*

# Types of Energy

## □ Examples of Transformation

| System             | From                  | To                                  | System              | From          | To                       |
|--------------------|-----------------------|-------------------------------------|---------------------|---------------|--------------------------|
| ATP Hydrolysis     | Chemical              | Mechanical                          | Heat Engine         | Thermal       | Mechanical               |
| Battery            | Chemical              | Electrical                          | Hydroelectric Dam   | Gravitational | Electrical               |
| Electric Generator | Kinetic<br>Mechanical | Electrical                          | Electric Lamp       | Electrical    | Thermal<br>Light         |
| Electric Heater    | Electrical            | Thermal                             | Microphone          | Sound         | Electrical               |
| Fire               | Chemical              | Thermal<br>Light                    | Ocean Thermal Power | Thermal       | Electrical               |
| Friction           | Kinetic               | Thermal                             | Photosynthesis      | Radiation     | Chemical                 |
| Fossil Fuel        | Chemical              | Thermal<br>Electrical<br>Mechanical | Piezoelectrics      | Strain        | Electrical               |
| Fuel Cell          | Chemical              | Electrical                          | Thermoelectric      | Thermal       | Electrical               |
| Geothermal Power   | Thermal               | Electrical                          | Wave Power          | Mechanical    | Electrical               |
|                    |                       |                                     | Windmill            | Wind          | Electrical<br>Mechanical |

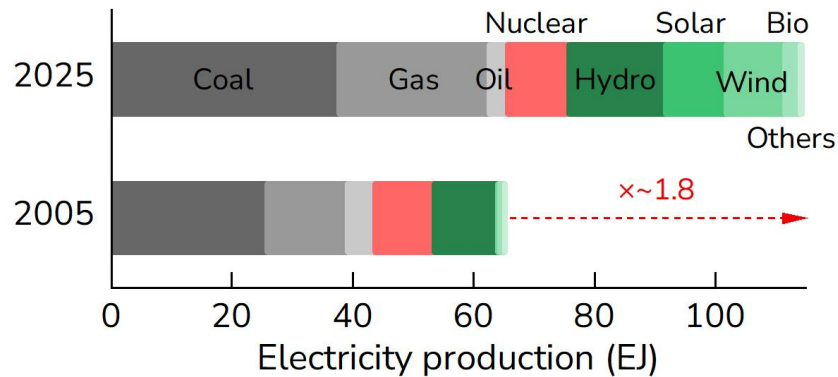
– There are many different systems, both natural and engineered

# Overview of Global Electricity

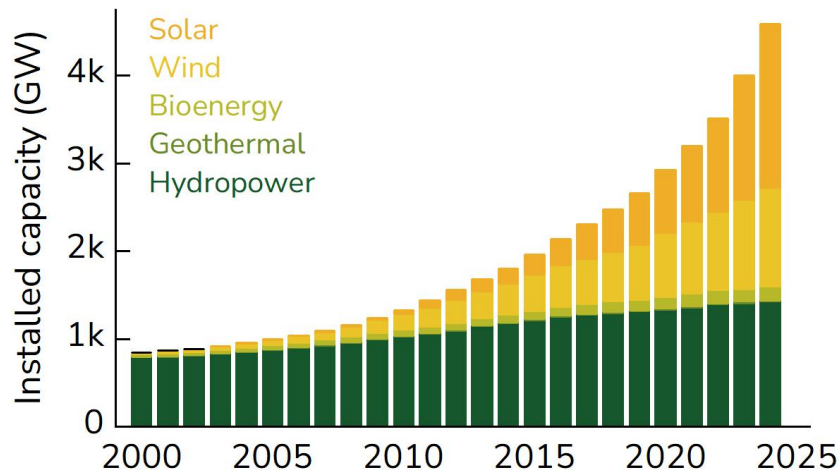
## Global Electricity Production

RE accounts for 34% in 2025 (18% in 2005)

\*Others: geothermal, biomass, waste, wave and tidal

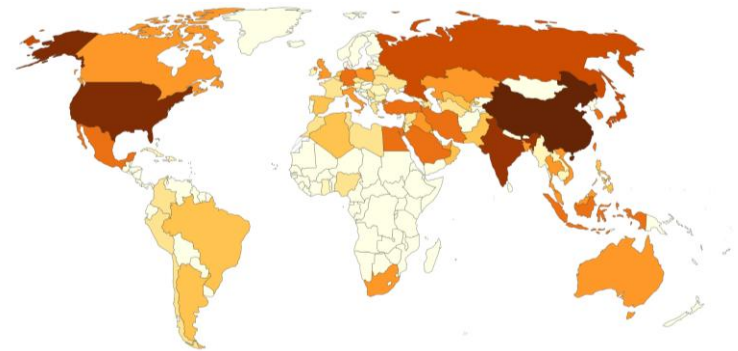


<Source: Statistical Review of World Energy (2025)>

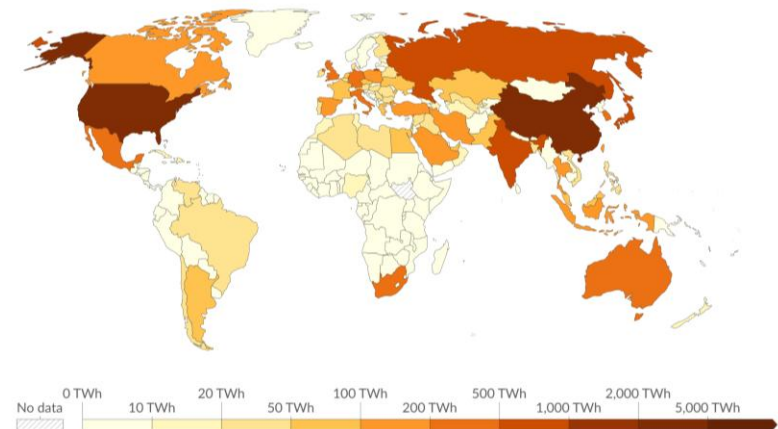


<Source: IRENA (2025)>

Electricity from Fossil Fuels (2025)



Electricity from Fossil Fuels (2005)



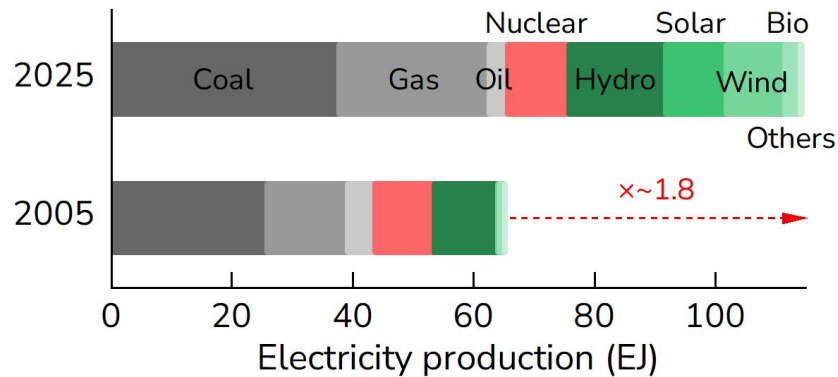
<Source: Our World in Data>

# Overview of Global Electricity

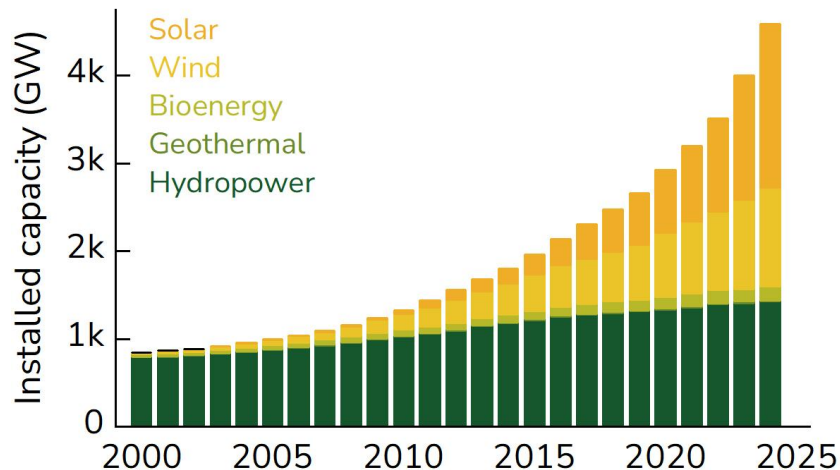
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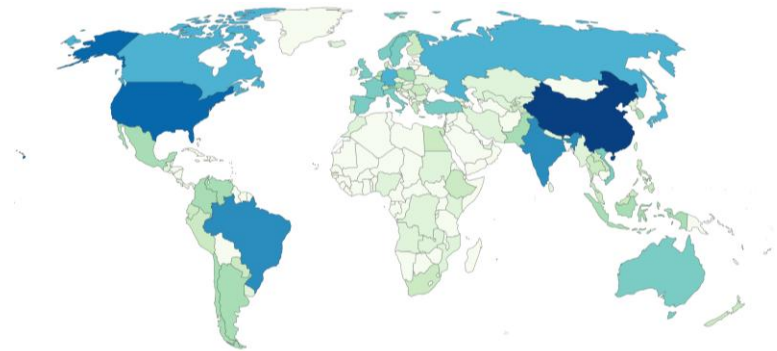


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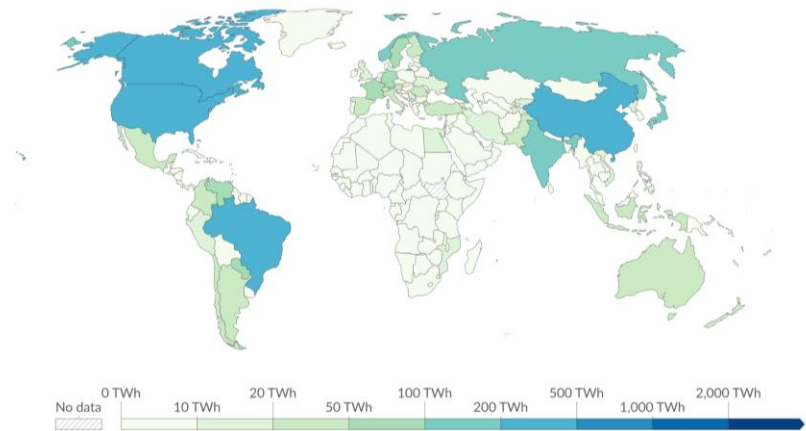


<Source: IRENA (2025)>

Electricity from Renewable (2025)



Electricity from Renewable (2005)



<Source: Our World in Data>

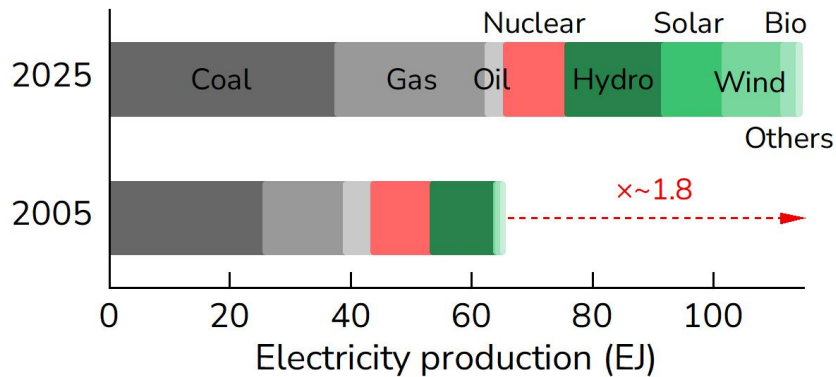
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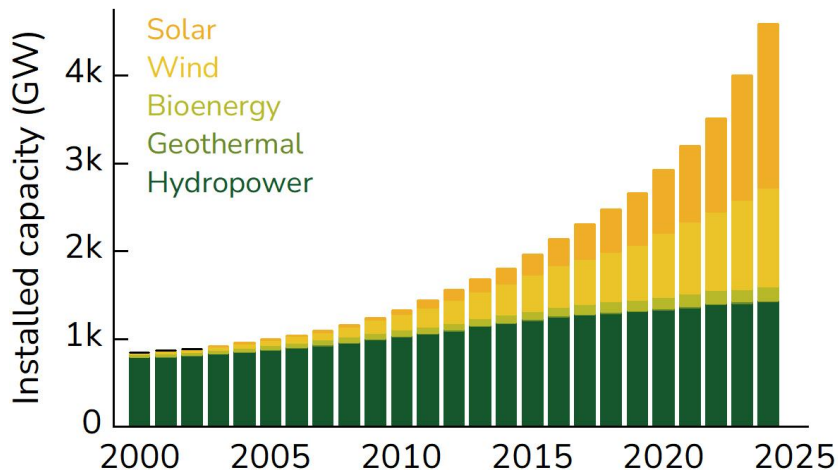
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\*Others: geothermal, biomass, waste, wave and tidal



<Source: Statistical Review of World Energy (2025)>

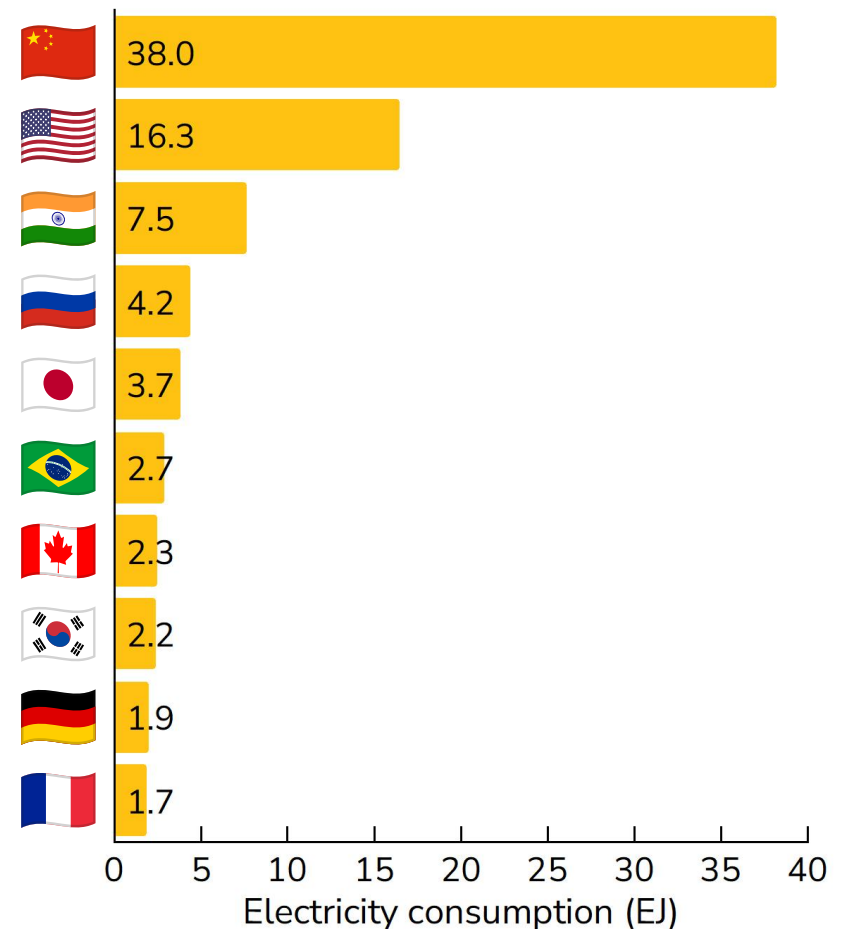


<Source: IRENA (2025)>

## Global Electricity Demand

Data centers consumed ~1.8 EJ in 2025

\*\*Usage of ~3.4 EJ in 2030



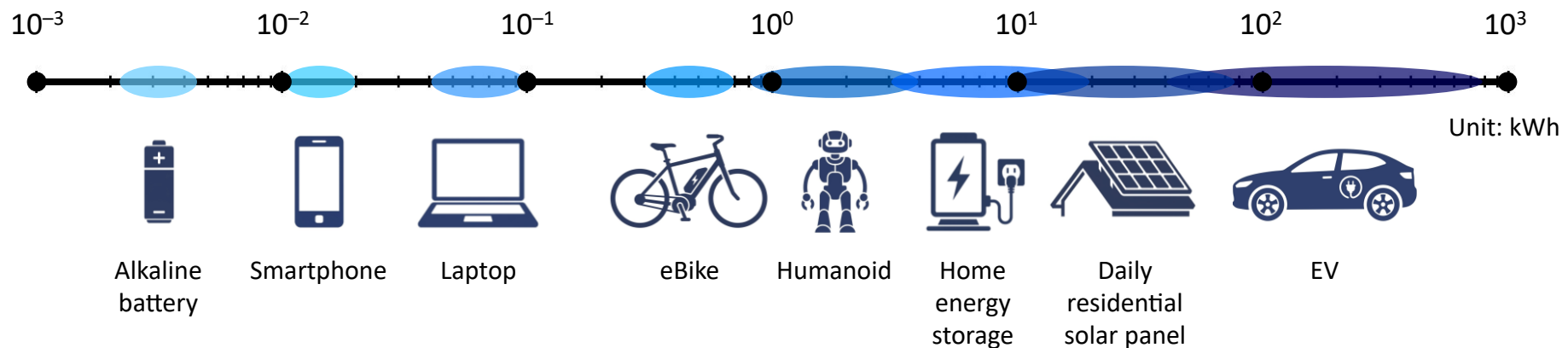
<Source: Ember (2026), Energy and AI (2026)>

# Scale for Electrical Energy

## □ What can 1 kWh (= $3.6 \times 10^6$ J = 3.6 MJ) do?

1 kWh = 1000 Wh; 1 kW (= 1000 W) of power can be used for an hour

- (EV) Drive a Tesla Model Y for ~6 km
- (TV) View an LG StanbyME 2 for ~43 h
- (Laptop) Power ~14 14-inch Macbook Pros
- (Smartphone) Fully charge ~70 iPhone 17s
- (Light Bulb) Operate a 12 W light bulb for ~83 h



# Scale for Electrical Energy

## ❑ What can 1 TWh ( $= 3.6 \times 10^{15} \text{ J} = 3.6 \text{ PJ}$ ) do?

1 TWh = 1,000,000,000,000 Wh = 1,000,000,000 kWh = 1,000,000 MWh = 1,000 GWh

- (EV) Drive around the world ~150,000 times in a Tesla Model Y
- (EV) Fully charge ~17,000,000 Tesla Model Ys
- (Home) Power ~300,000 Korean households for a year
- (Industry) Run Samsung for 2 weeks (Annual consumption: ~26 TWh))
- (City) Power the entire city of Seoul for a week (Annual consumption: ~50 TWh)

# Alternative Energy Sources

## ■ Solar

- Solar Thermal Energy  
(Thermal process)
- Photovoltaics (PV)  
(Quantum process)

- Ultimate source of most energy
- Vast potential
- Low efficiency
- Limited upscaling, distribution

## ■ Nuclear

- Fission, Fusion  
(Quantum process)

- High power output
- Vast potential
- Toxic waste
- Disaster risk

## ■ Others

- Wind, Hydro, Wave  
(Mechanical process)
- Ocean, Geothermal  
(Thermal process)

- Clean energy sources
- Substantial potential
- Intermittency
- Geographic limitation



# Alternative Energy Sources

## ❑ Drawbacks of Renewables

Compared to nuclear power, renewables offer lower LCOE but exhibit lower capacity factors

- **Levelised Cost of Electricity, *LCOE***

*LCOE* = discounted sum of costs over lifetime / discounted sum of electricity over lifetime

- **Capacity Factor, *CF***

$$CF = E_{\text{total}} / (P_n \times t)$$

( $E_{\text{total}}$  : total electricity over the time period  $t$ )

$P_n$  : installed capacity of the device or system)

(e.g.) Three Gorges Dam (China) is, with its installed capacity of 22,500 MW (2017), the largest power generating station in the world by installed capacity. In 2015 it generated 87 TWh, for a capacity factor of **87 TWh yr<sup>-1</sup> / 22500 MW = 45%**

(e.g.) Agua Caliente Solar Project (AZ, US) has an installed capacity of 290 MW and an average annual production of 740 GWh yr<sup>-1</sup>. Its capacity factor is **= %**

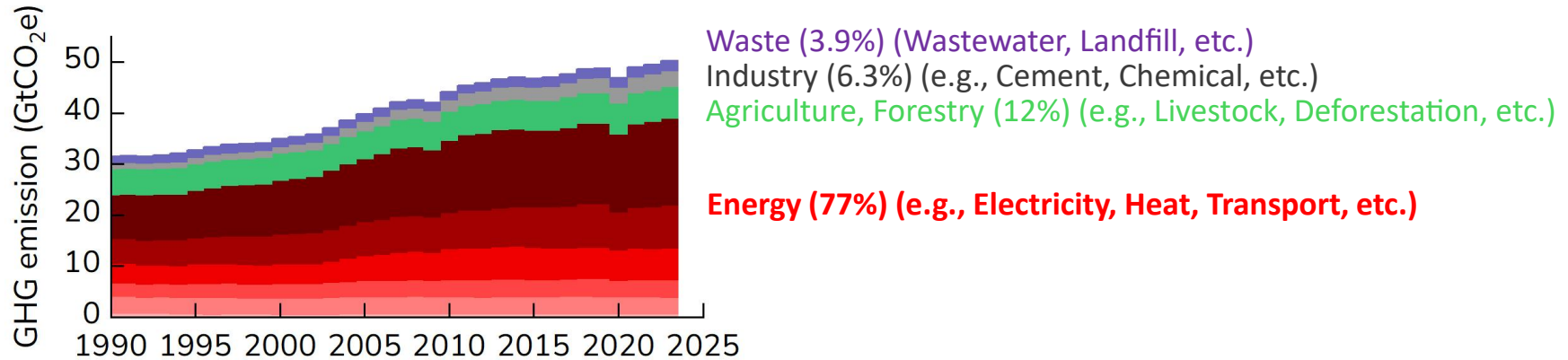
|                                       | Solar | Wind<br>(Onshore) | (Offshore) | Hydropower | Geothermal | Biomass | Nuclear |
|---------------------------------------|-------|-------------------|------------|------------|------------|---------|---------|
| Installed Cost (\$ kW <sup>-1</sup> ) | 691   | 1,041             | 2,852      | 2,267      | 4,015      | 3,242   | ~5,500  |
| LCOE (\$ MWh <sup>-1</sup> )          | 43    | 34                | 79         | 57         | 60         | 87      | ~75     |
| CF (%)                                | 17    | 34                | 42         | 48         | 88         | 73      | ~90     |

# Why Renewables?

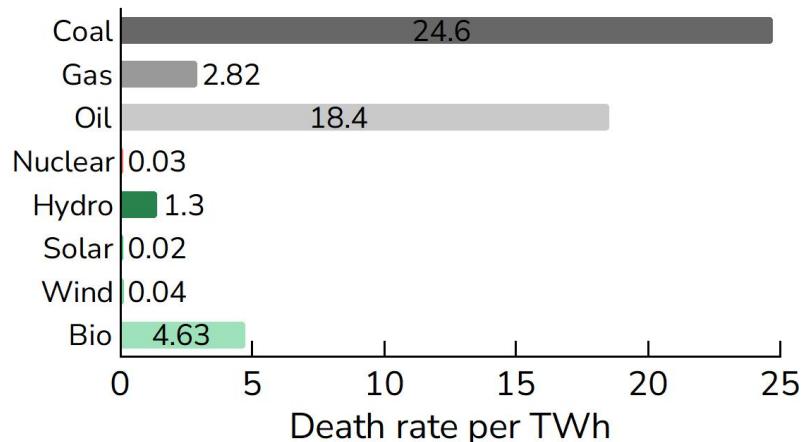


## Global GHG Emissions by Sector

Greenhouse gas emissions in 2023 were ~50 billion tonnes CO<sub>2</sub>eq, with ~80% coming from *energy*

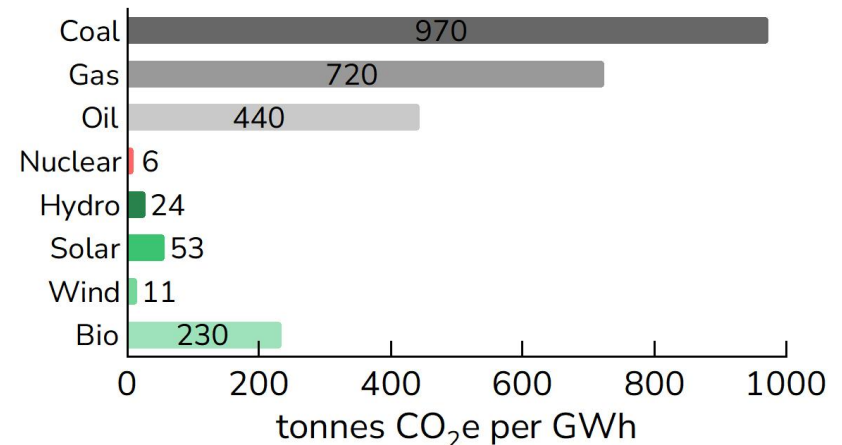


## Air Pollution & Accident



<Source: Our World in Data>

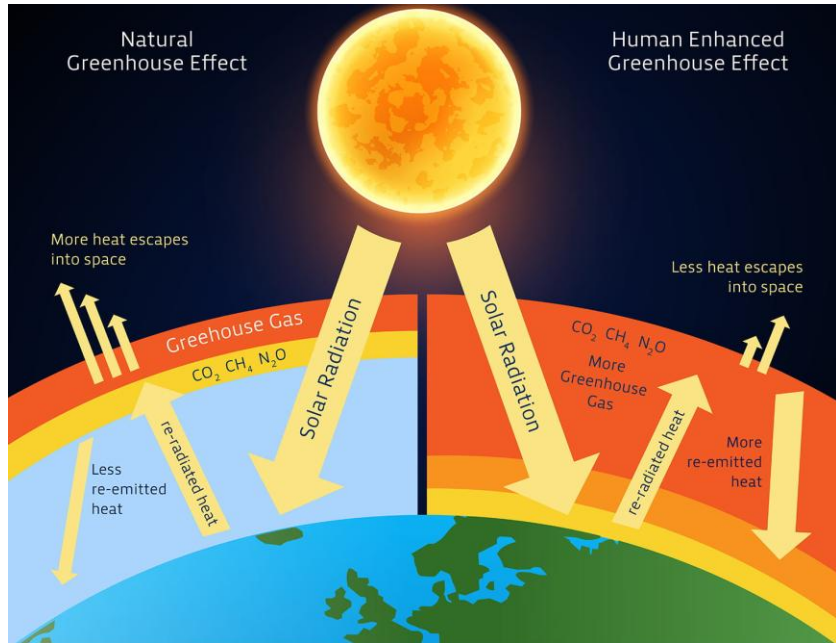
## GHG Emissions



# Greenhouse Effect

## Greenhouse Effect

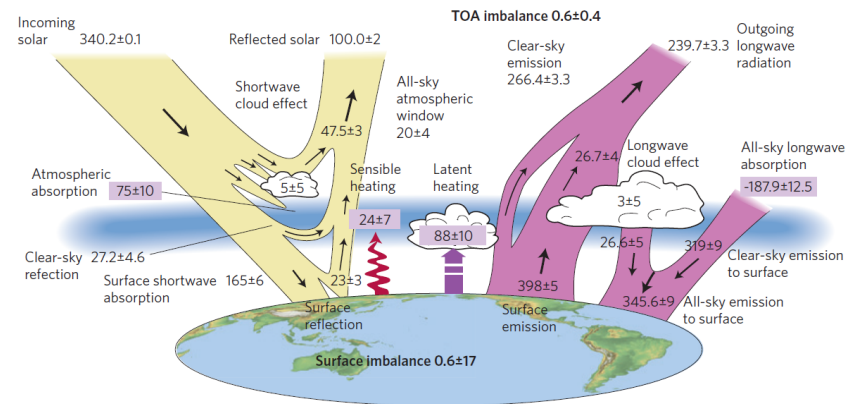
GHGs prevent the planet from losing heat to space



*Renew. Sustain. Energy Rev.*, **71** (2017) 112–126

## Earth's Energy Budget

Annual Earth Energy Imbalance =  $\sim 0.6 \text{ W m}^{-2}$



**Fig.** Global annual mean energy budget of Earth for the approximate period 2000–2010.

Yellow: Solar fluxes

Pink: Infrared fluxes

*Nat. Geosci.*, **5** (2012) 691–696

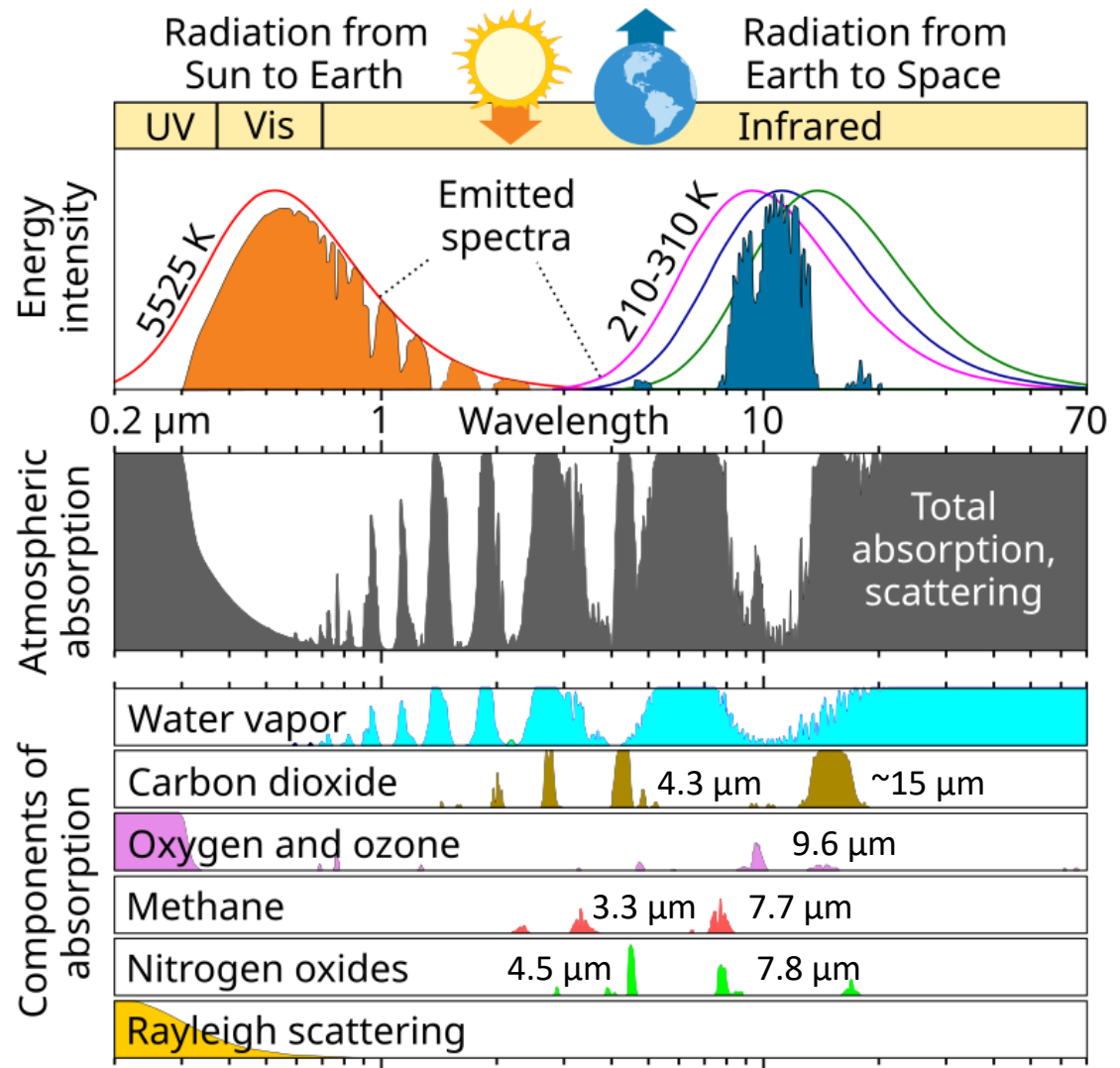
- The surface of the Sun is  $\sim 5500^\circ\text{C}$ , so it emits *shortwave* radiation (UV, VL, IR)
- Shortwave radiation passes through GHGs and heats the Earth's surface
- The Earth's surface emits *longwave* radiation that is mostly absorbed by GHGs

# Greenhouse Effect



## □ Atmosphere of Earth

| Gas                   | Volume fraction (ppm) |
|-----------------------|-----------------------|
| N <sub>2</sub>        | 780,840               |
| O <sub>2</sub>        | 209,460               |
| Ar                    | 9,340                 |
| <b>CO<sub>2</sub></b> | <b>420</b>            |
| Ne                    | 18.182                |
| He                    | 5.24                  |
| <b>CH<sub>4</sub></b> | <b>1.92</b>           |
| Kr                    | 1.14                  |
| H <sub>2</sub>        | 0.55                  |
| <b>N<sub>2</sub>O</b> | <b>0.33</b>           |
| <b>CO</b>             | <b>0.10</b>           |
| Xe                    | 0.09                  |
| <b>O<sub>3</sub></b>  | <b>0.07</b>           |
| <b>NO<sub>2</sub></b> | <b>0.02</b>           |
| <b>H<sub>2</sub>O</b> | <b>Up to 40,000</b>   |



<Source: National Oceanic and Atmospheric Administration>

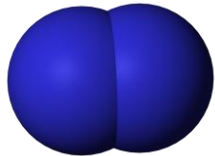
# Greenhouse Effect



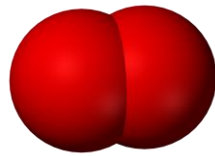
## ☐ Infrared Inactive Molecules

Gases with single or two identical atoms

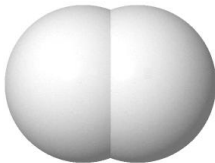
Nitrogen ( $\text{N}_2$ )



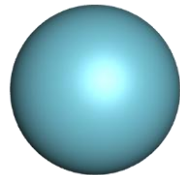
Oxygen ( $\text{O}_2$ )



Hydrogen ( $\text{H}_2$ )



Helium ( $\text{He}$ )

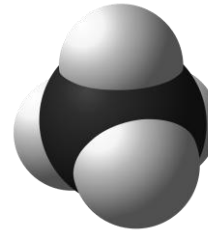


- Transparent to longwave radiation
- No net change in dipole moment

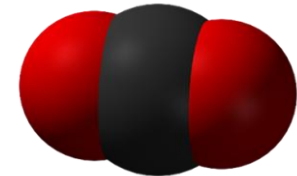
## ☐ Infrared Active Molecules

Gases with two or more different atoms

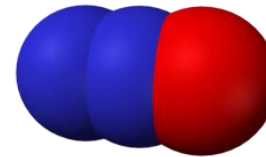
Methane ( $\text{CH}_4$ )



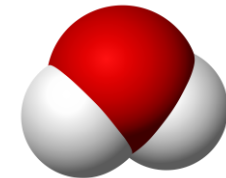
Carbon dioxide ( $\text{CO}_2$ )



Nitrous oxide ( $\text{N}_2\text{O}$ )



Water/Steam ( $\text{H}_2\text{O}$ )



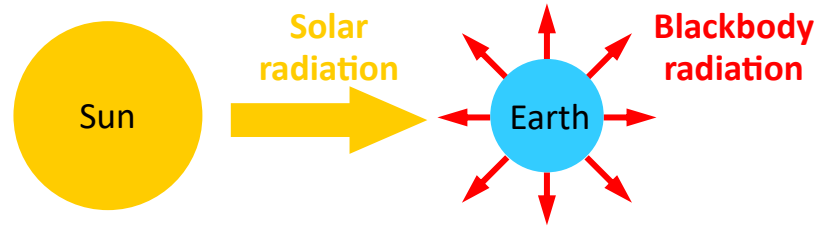
- Absorbing IR radiation in the bonds
- Vibrations modify the dipole moment

# Greenhouse Effect



## What if...

Without GHGs, the average surface temperature of the Earth would be as cold as  $\sim -19^\circ\text{C}$



Under the black-body approximation, the Sun's luminosity  $L_\odot$  is

$$L_\odot = 4\pi R_\odot^2 \sigma T_\odot^4 \quad (1)$$

, where  $R_\odot$  and  $T_\odot$  are respectively the radius and temperature of the Sun and  $\sigma$  is the Stefan–Boltzmann constant.

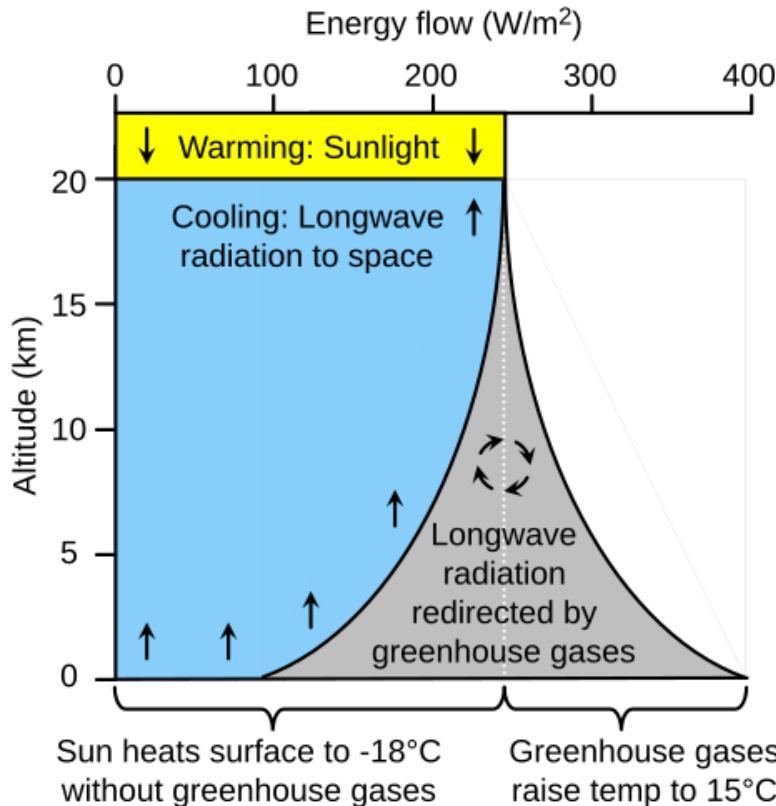
This energy is passing through a sphere with a radius of  $a_0$ , the distance between the Earth and the Sun (i.e., 1 AU), and the irradiance (received power per unit area)  $E_\oplus$  is

$$E_\oplus = \frac{L_\odot}{4\pi a_0^2} \quad (2)$$

The radiant flux (i.e. solar power) absorbed by the Earth is

$$\Phi_{\text{abs}} = 4\pi R_\oplus^2 Q_0 = \pi R_\oplus^2 E_\oplus \quad (3)$$

, where  $R_\oplus$  is the radius of the Earth and  $Q_0$  is global mean solar irradiance.

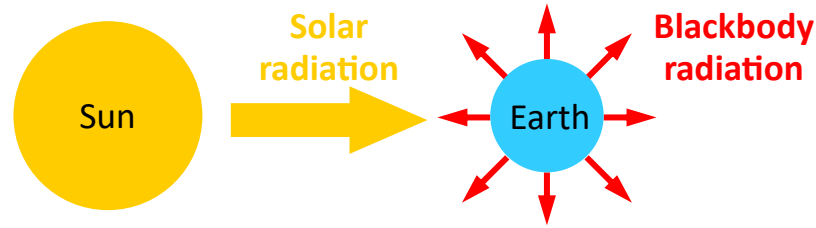


# Greenhouse Effect



## What if...

Without GHGs, the average surface temperature of the Earth would be as cold as  $\sim -19^\circ\text{C}$



Because the Stefan–Boltzmann law uses a fourth power, it has a stabilizing effect on the exchange, and the flux emitted by Earth tends to be equal to the flux absorbed, close to the steady state

$$4\pi R_{\oplus}^2 \sigma T_{\oplus}^4 = \pi R_{\oplus}^2 E_{\oplus} \quad (4)$$

$$= \pi R_{\oplus}^2 \frac{4\pi R_{\odot}^2 \sigma T_{\odot}^4}{4\pi a_0^2} \quad (5)$$

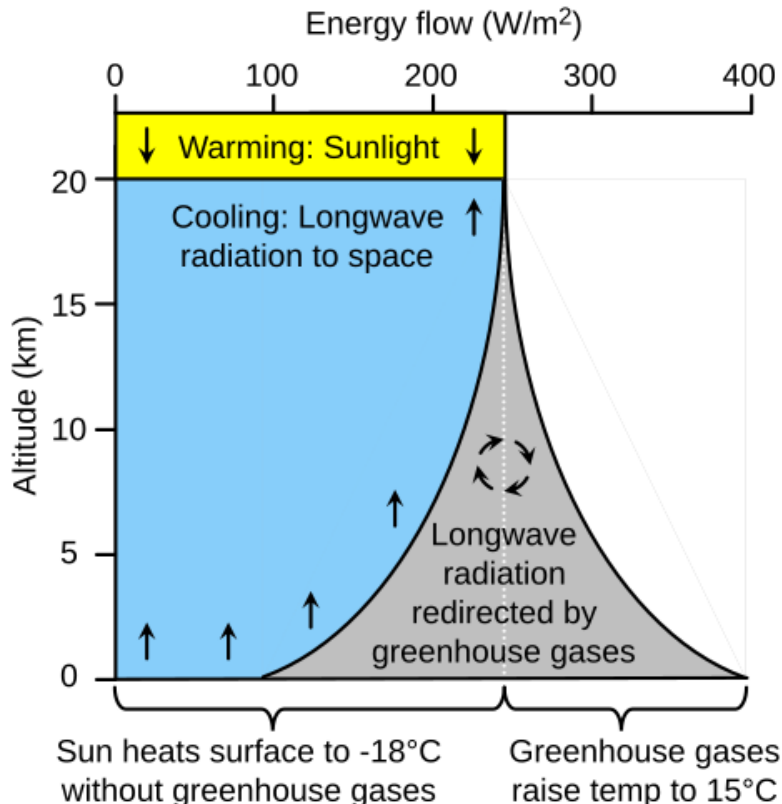
The effective temperature of the Earth  $T_{\oplus}$  can be expressed by

$$T_{\oplus}^4 = \frac{R_{\odot}^2 T_{\odot}^4}{4a_0^2} \Leftrightarrow T_{\oplus} = T_{\odot} \sqrt{\frac{R_{\odot}}{2a_0}} \quad (6)$$

Since  $R_{\odot} = 6.957 \times 10^8 \text{ m}$ ,  $T_{\odot} = 5780 \text{ K}$ , and  $a_0 = 1.496 \times 10^{11} \text{ m}$ ,  $T_{\oplus}$  is estimated as to be  $278.7 \text{ K}$  ( $=5.564^\circ\text{C}$ ), assuming that it perfectly absorbs all emission falling on it and has no atmosphere.

The Earth has an albedo  $\alpha$  of 0.306, reducing the temperature by a factor of  $0.694^{1/4}$ , giving **254.4 K ( $=-18.76^\circ\text{C}$ )**.

Because of the greenhouse effect, the Earth's actual average surface temperature is  **$\sim 287.2 \text{ K}$  ( $\sim 14.1^\circ\text{C}$ )**, which is higher than the estimated effective temperature, and even higher than the  $297 \text{ K}$  temperature that a blackbody would have.

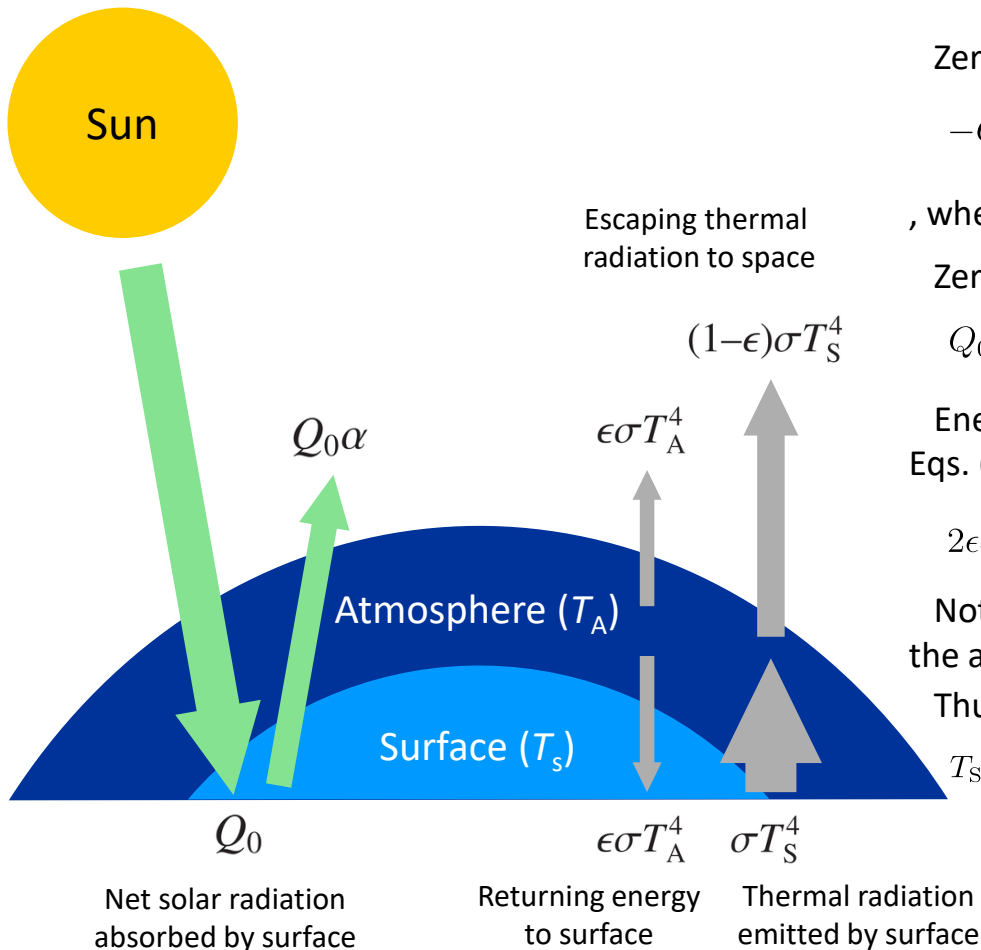


# Greenhouse Effect



## □ Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



Zero net radiation leaving the top of the atmosphere requires

$$-Q_0(1 - \alpha) + \epsilon\sigma T_A^4 + (1 - \epsilon)\sigma T_S^4 = 0 \quad (7)$$

, where  $\epsilon$  is the emissivity of the atmosphere.

Zero net radiation entering the surface requires

$$Q_0(1 - \alpha) + \epsilon\sigma T_A^4 - \sigma T_S^4 = 0 \quad (8)$$

Energy equilibrium of the atmosphere can be derived from Eqs. (7) and (8),

$$2\epsilon\sigma T_A^4 - \epsilon\sigma T_S^4 = 0 \quad (9)$$

Note the important factor of 2, resulting from the fact that the atmosphere radiates both upward and downward.

Thus, the ratio of  $T_A$  to  $T_S$  is independent of  $\epsilon$

$$T_S = 2^{0.25} T_A = 1.189 T_A \quad (10)$$

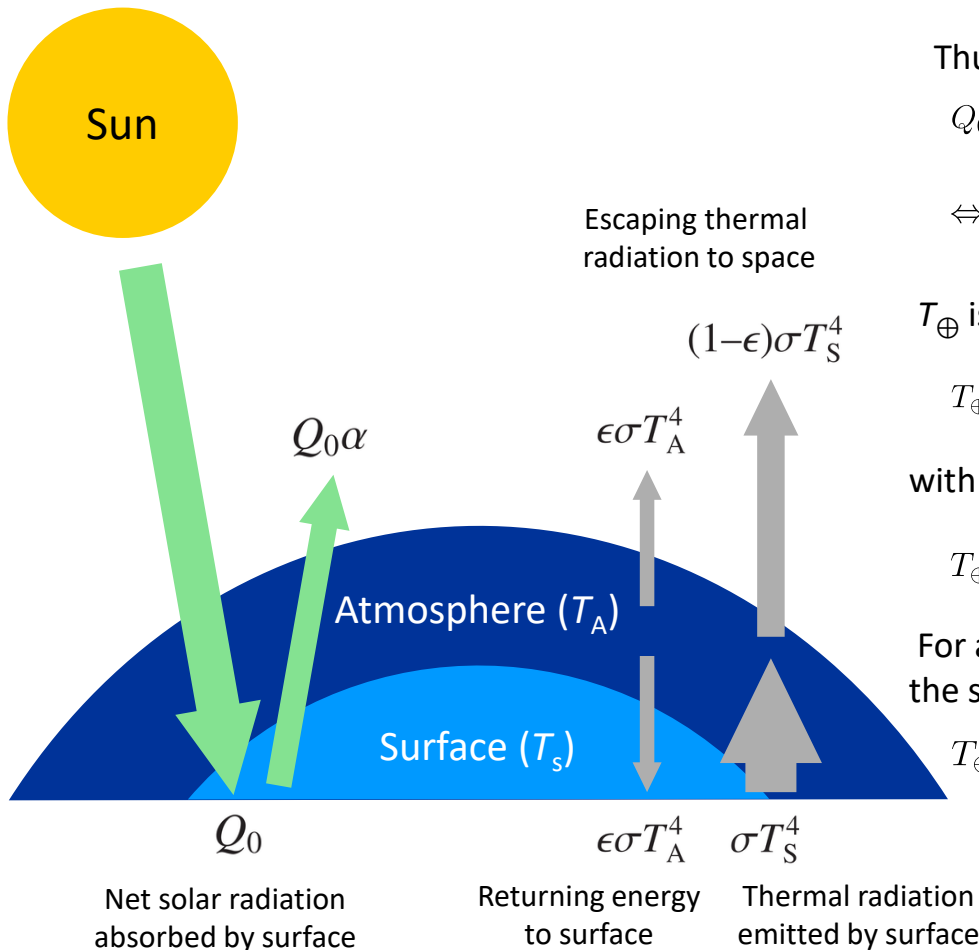


# Greenhouse Effect



## □ Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



Thus,  $T_A$  can be expressed in terms of  $T_s$ , and then Eq. (8) gives

$$Q_0(1 - \alpha) = (1 - \frac{\epsilon}{2})\sigma T_s^4 \quad (11)$$

$$\Leftrightarrow T_s = \left[ \frac{Q_0(1 - \alpha)}{\sigma} \frac{1}{1 - \frac{\epsilon}{2}} \right]^{\frac{1}{4}} \quad (12)$$

$T_{\oplus}$  is the solution for  $T_s$ , for the case of  $\epsilon = 0$  (no atmosphere)

$$T_{\oplus} = \left[ \frac{Q_0(1 - \alpha)}{\sigma} \right]^{\frac{1}{4}} \quad (13)$$

with the definition of  $T_{\oplus}$ :

$$T_{\oplus} = T_s \left[ \frac{1}{1 - \frac{\epsilon}{2}} \right]^{\frac{1}{4}} \quad (14)$$

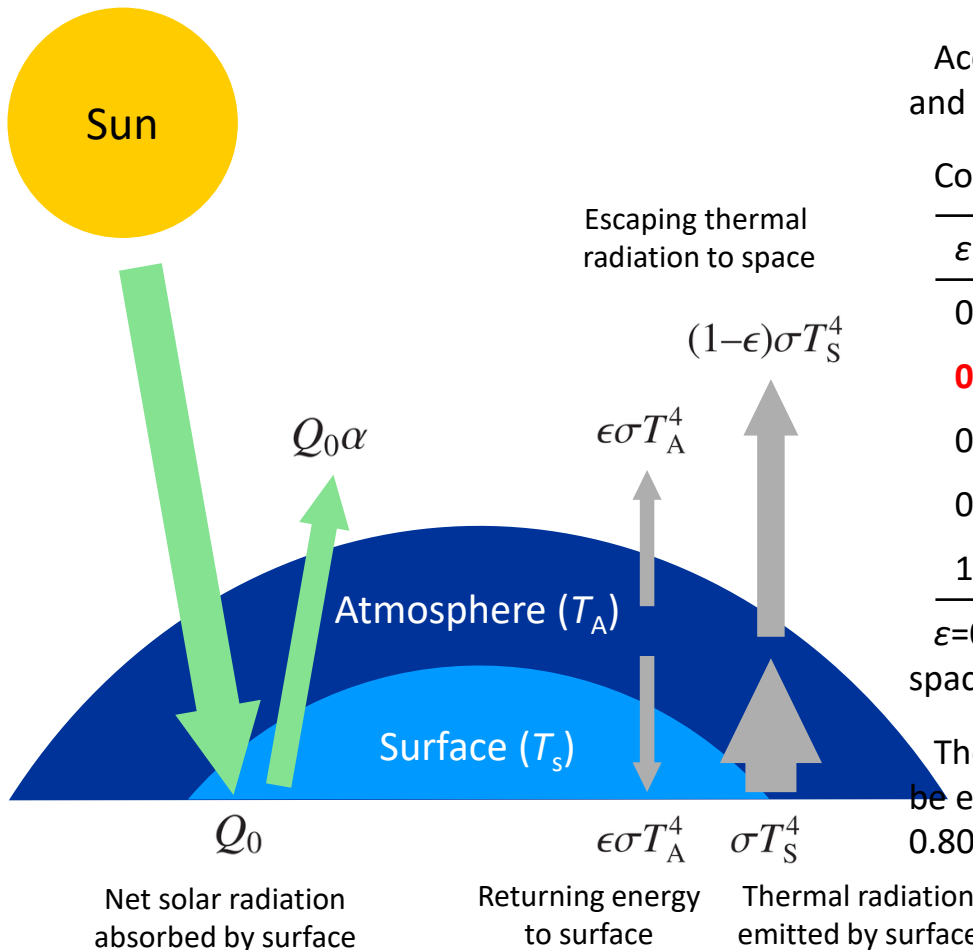
For a perfect greenhouse, with no radiation escaping from the surface, or  $\epsilon=1$ :

$$T_{\oplus} = T_A \quad (15)$$

# Greenhouse Effect

## ☐ Idealized Greenhouse Model

A simple model for mechanistically understanding the greenhouse effect on Earth



According to Eqs. (1) and (2),  $E_{\oplus}$  is estimated as  $1369 \text{ W m}^{-2}$ , and thus  $Q_0$  is  $342.2 \text{ W m}^{-2}$  ( $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ )

Considering  $\alpha$  is 0.306,  $T_S$  can be calculated as below (Eq. (12)):

| $\epsilon$ (-) | $T_S$ (K)    | $T_S$ ( $^{\circ}\text{C}$ ) |
|----------------|--------------|------------------------------|
| 0              | 254.4        | -18.76                       |
| <b>0.78</b>    | <b>287.8</b> | <b>14.70</b>                 |
| 0.80           | 289.0        | 15.89                        |
| 0.82           | 290.3        | 17.11                        |
| 1              | 302.5        | 29.37                        |

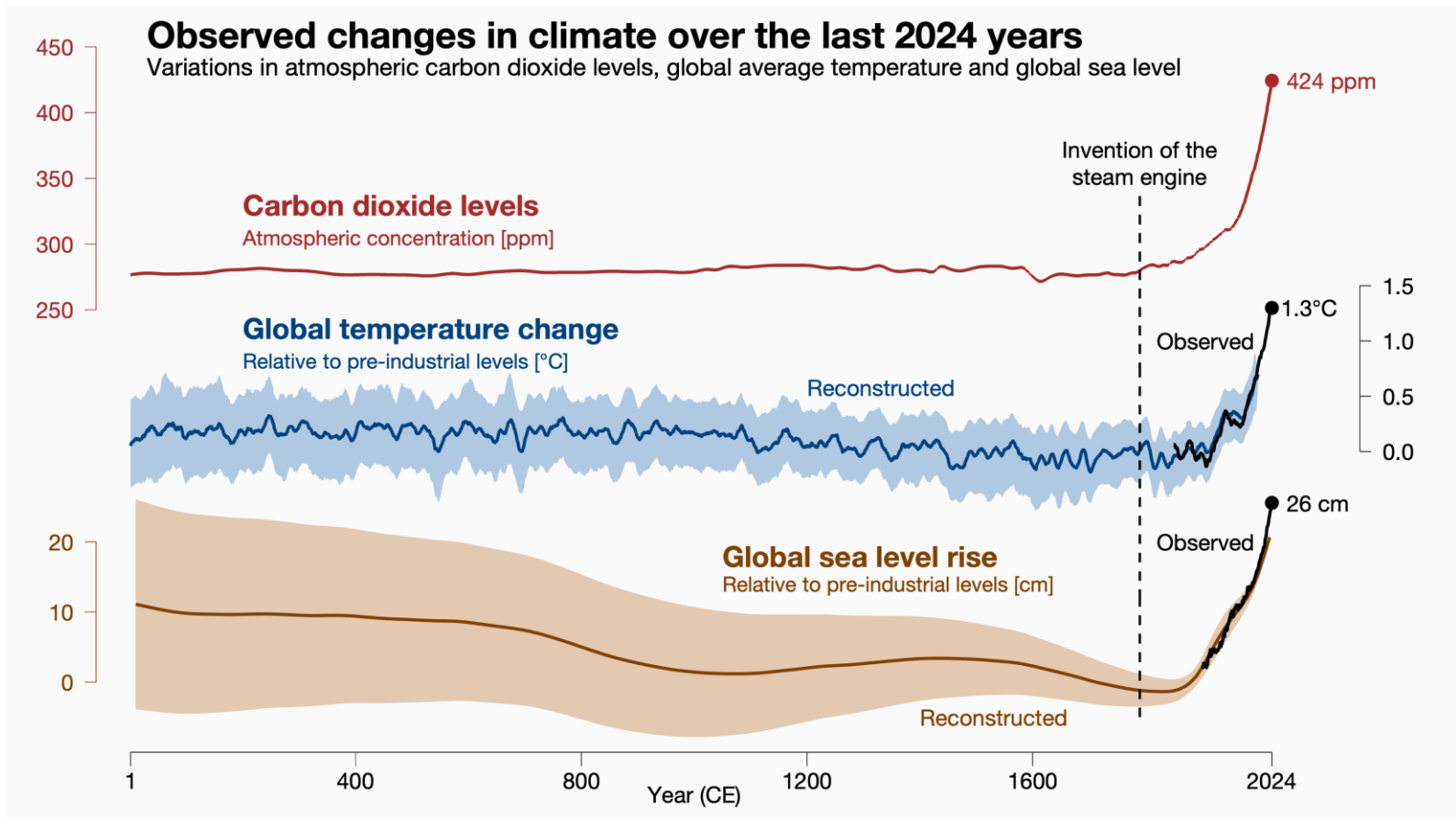
$\epsilon=0.78$  implies 22% of the surface radiation escapes directly to space.

The radiative forcing for doubling  $\text{CO}_2$  is  $\sim 3.71 \text{ W m}^{-2}$ , which can be estimated by using  $\Delta\epsilon = 0.019$ . Thus, a change of  $\epsilon$  from 0.78 to 0.80 is consistent with the radiative forcing from a doubling of  $\text{CO}_2$ .

# Greenhouse Effect



## □ Atmospheric CO<sub>2</sub> and Global Temperature

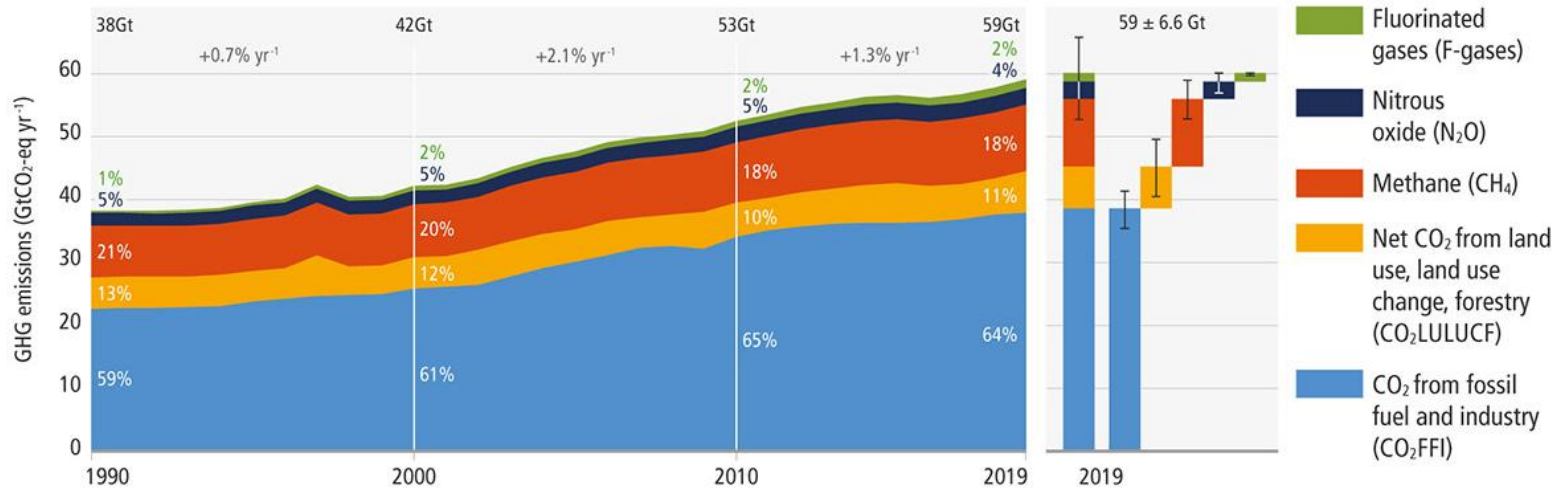


<Source: Climate Visuals>

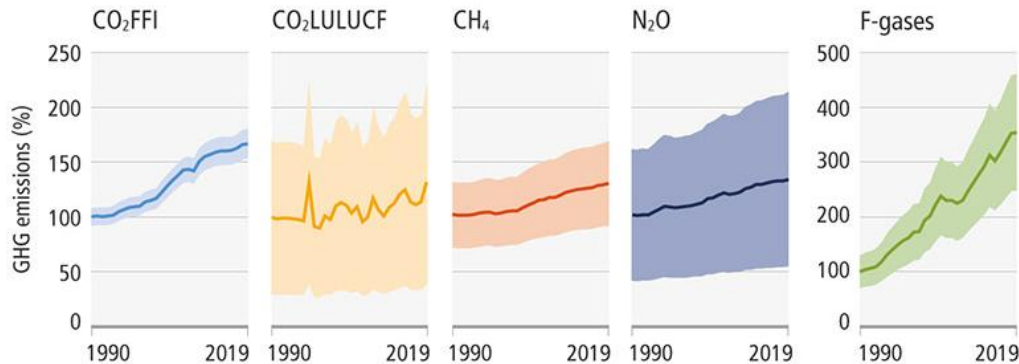
# Greenhouse Effect

## Global Net Anthropogenic GHG Emissions

a. Global net anthropogenic GHG emissions 1990–2019 <sup>(5)</sup>



b. Global anthropogenic GHG emissions and uncertainties by gas – relative to 1990



|                        | 2019 emissions (GtCO <sub>2</sub> -eq) | 1990–2019 increase (GtCO <sub>2</sub> -eq) | Emissions in 2019, relative to 1990 (%) |
|------------------------|----------------------------------------|--------------------------------------------|-----------------------------------------|
| CO <sub>2</sub> FFI    | 38±3                                   | 15                                         | 167                                     |
| CO <sub>2</sub> LULUCF | 6.6±4.6                                | 1.6                                        | 133                                     |
| CH <sub>4</sub>        | 11±3.2                                 | 2.4                                        | 129                                     |
| N <sub>2</sub> O       | 2.7±1.6                                | 0.65                                       | 133                                     |
| F-gases                | 1.4±0.41                               | 0.97                                       | 354                                     |
| Total                  | 59±6.6                                 | 21                                         | 154                                     |

The solid line indicates central estimate of emissions trends. The shaded area indicates the uncertainty range.

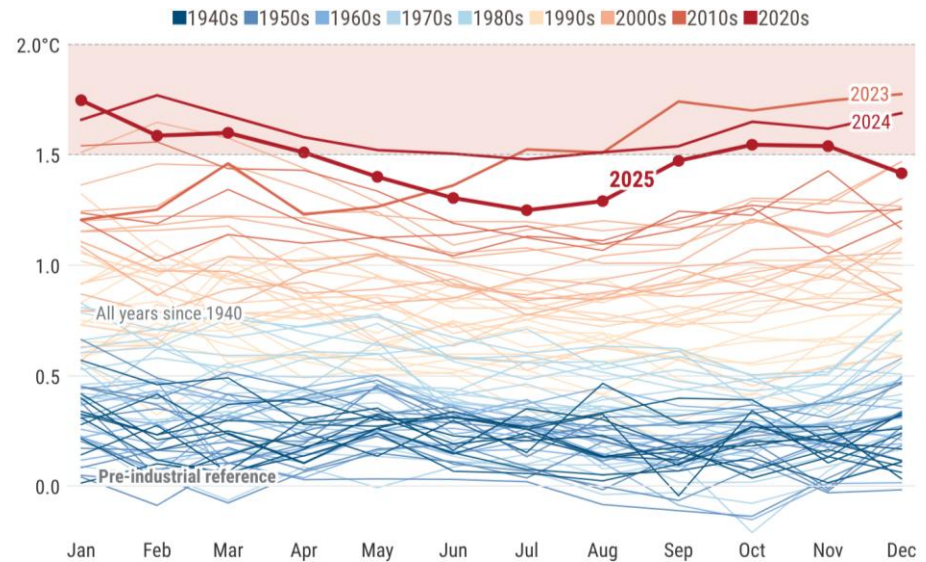
<Source: IPCC>

# Climate Change

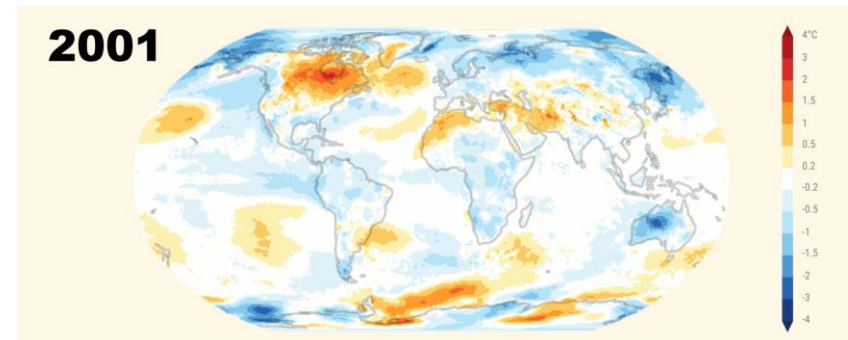
## Global Boiling



Monthly global surface air temperature anomalies



Annual surface air temperature anomalies (2001–2025)



<Source: Copernicus Climate Change (2025)>



# Climate Change-Driven Disasters

## ☐ Heatwave



Hajj extreme heat disaster (Jun 2024)

## ☐ Tropical Cyclone



Hurricane Helene (Sep 2024)

## ☐ Wildfire



LA wildfire (Jan 2025)

## ☐ Drought



Amazon deforestation

## ☐ Flood



Spanish floods (Oct–Nov 2024)

## ☐ Particulate matter

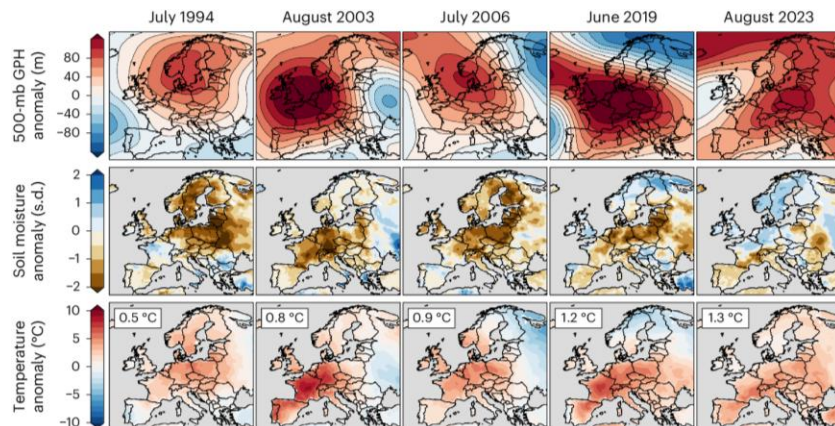


Air pollution in London

# Climate Change-Driven Disasters

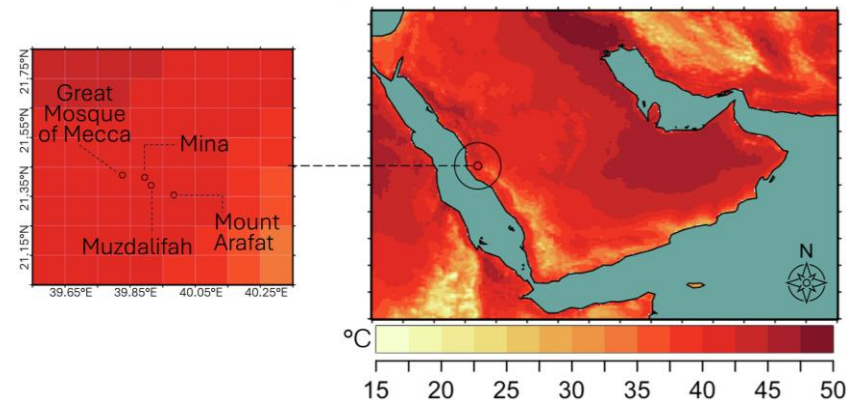
## European Heatwaves

**More than 62,000 people died in Europe's 2024 heatwave: Which country was hit the hardest?**



## Hajj Extreme Heat Disaster

**At least 1,301 people died during Hajj - Saudi Arabia**

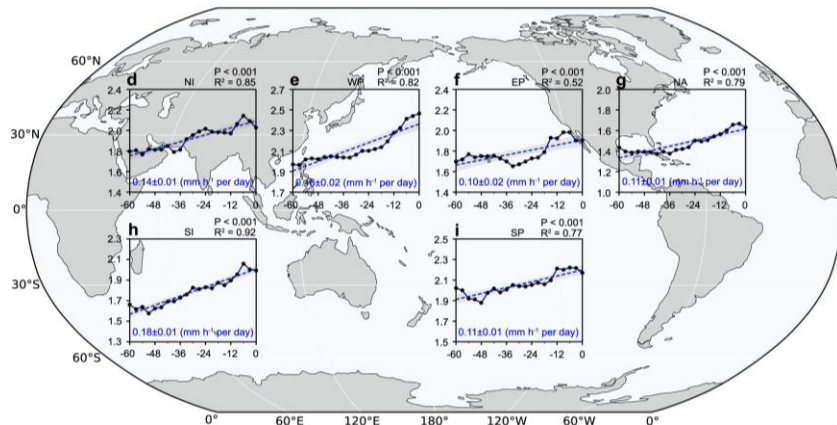




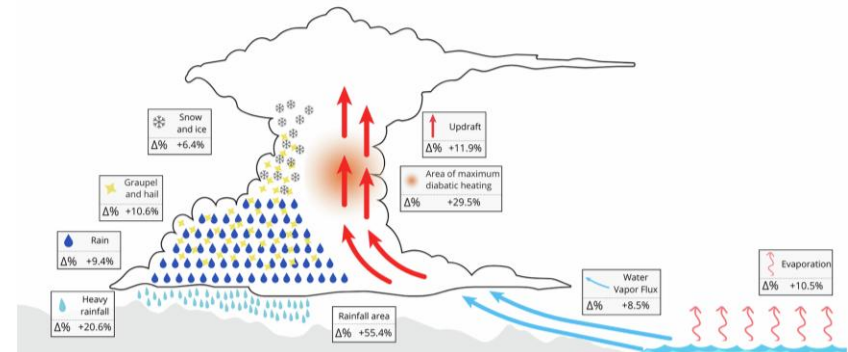
# Climate Change-Driven Disasters

## ☐ Tropical Cyclone

### Ferocity of Atlantic hurricanes surges as the ocean warms



## ☐ Heavy Rainfalls

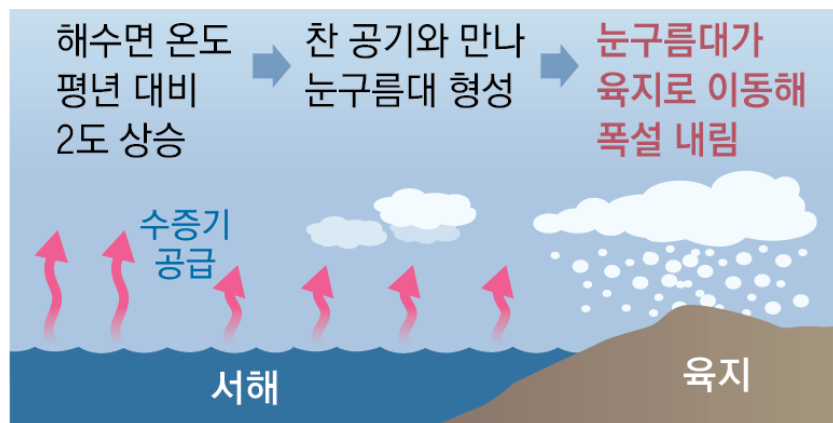




# Climate Change-Driven Disasters

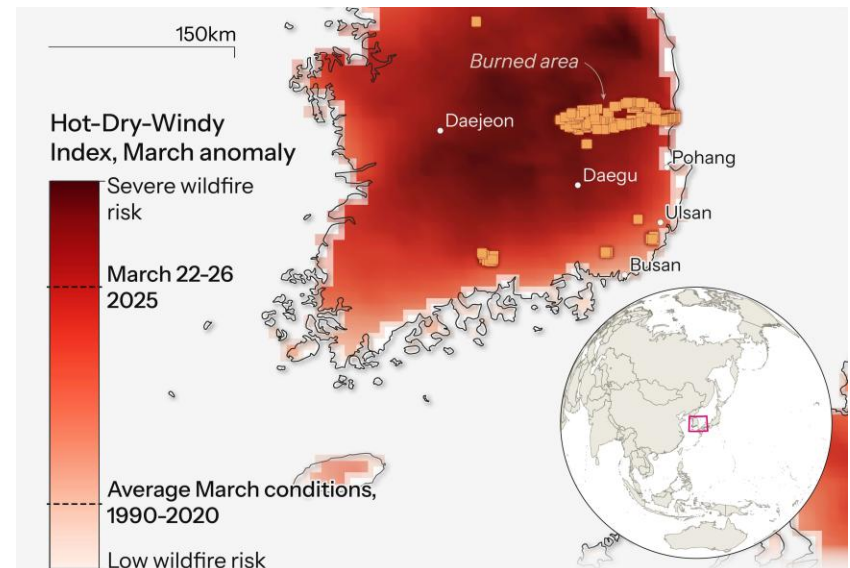
## ❑ Snowstorm

117년 만 11월 폭설, 서울 20cm 넘는 눈 처음



## ❑ Wildfire

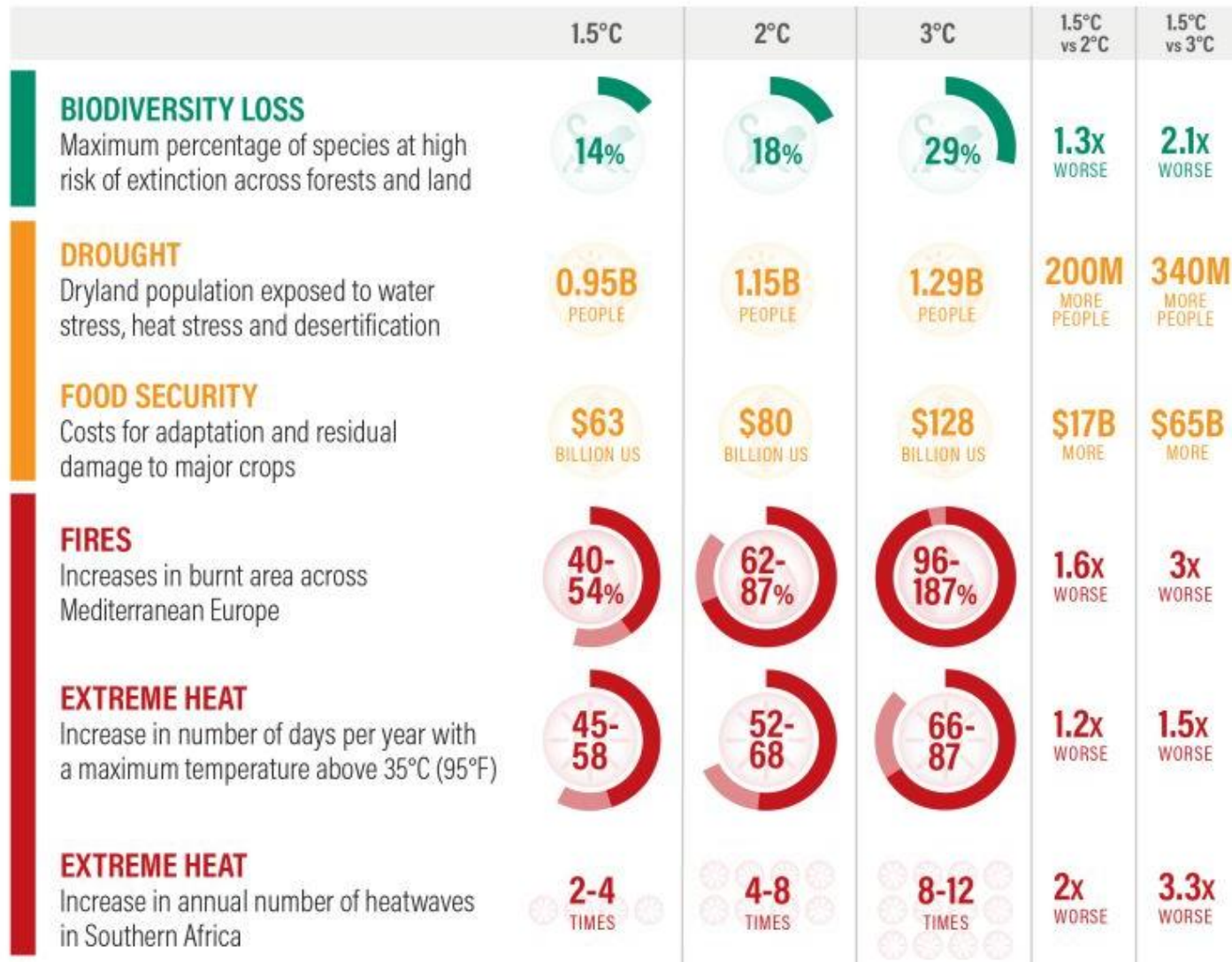
영남 산불 피해면적 10만4천ha로 잠정 집계



<Source: World Weather Attribution (WWA)>

# Climate Impacts

## Temperature Rise Scenarios and Climate Change Impacts



# Climate Impacts

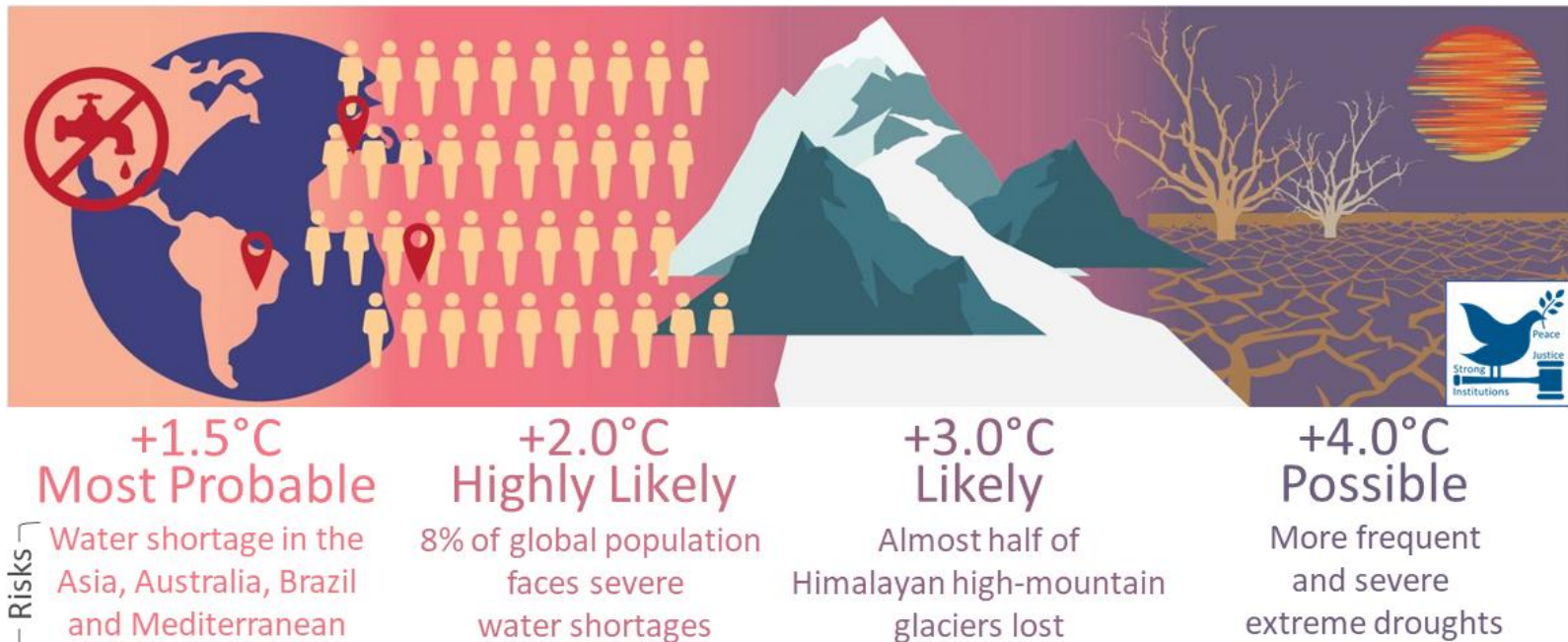
## □ Temperature Rise Scenarios and Climate Change Impacts (Cont'd)



<Source: Climate Resources Institute>

# Climate Impacts

## ☐ Temperature Rise Scenarios and Climate Change Impacts



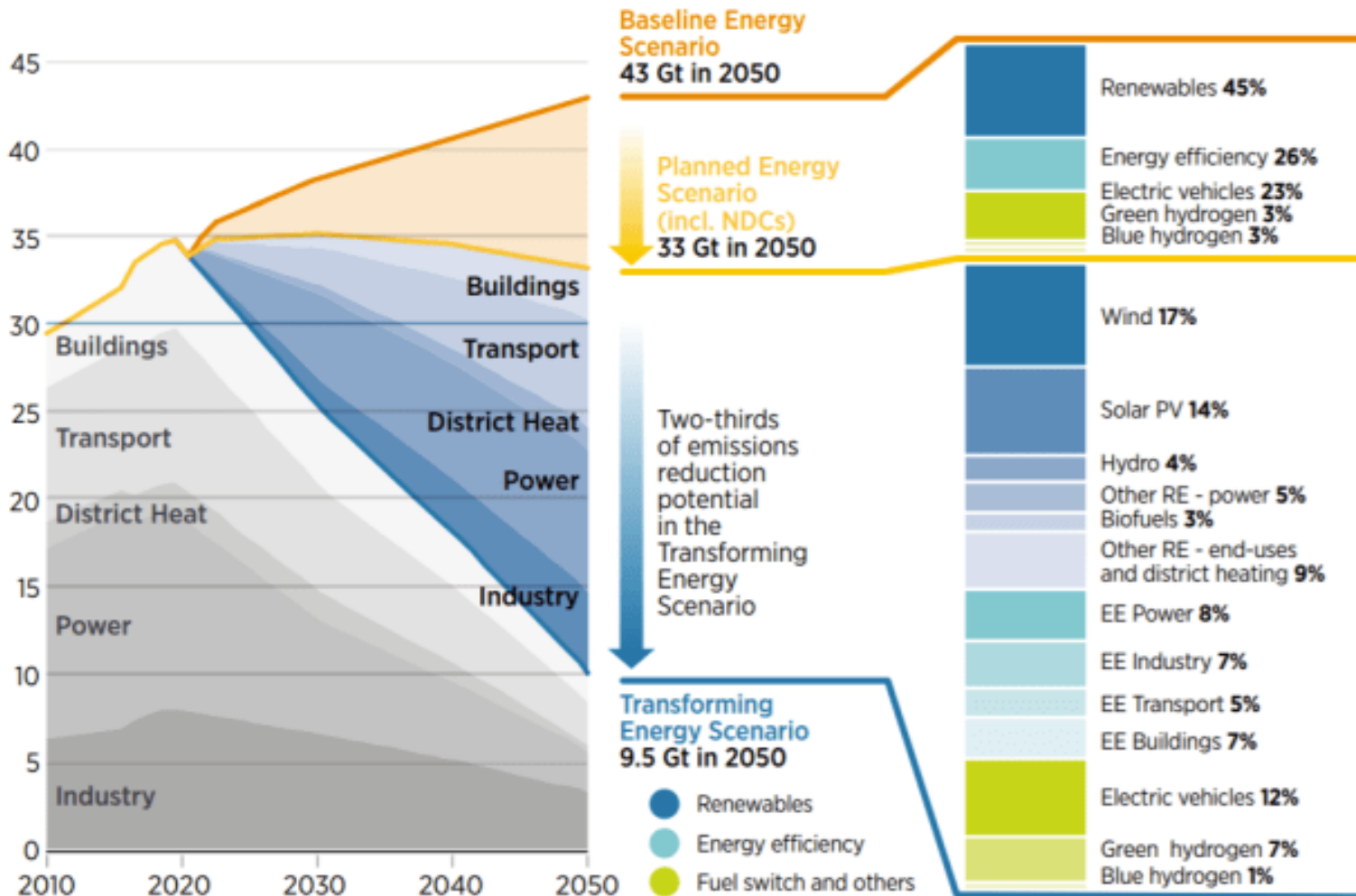
Risks

<Source: IPCC>



# Pathways to 1.5 °C

## □ Mitigating GHG Emissions by Energy Transition



# Assignment (~08<sup>th</sup> Sep)

## □ Planetary Equilibrium Temperature

Consider a planetary body orbiting the Sun at a distance  $a$  with an albedo  $\alpha$  and radius  $R$ .

The Sun has a luminosity  $L_{\odot}=4\pi R_{\odot}^2\sigma T_{\odot}^4$ , where  $R_{\odot}$  is the radius and  $T_{\odot}$  is the effective surface temperature of the Sun. Solve the following problems:

- Express the solar irradiance  $E$  (received power per unit area) at distance  $a$ , and write down the net solar radiation  $Q_0$  entering the planetary surface using  $\sigma$ ,  $T_{\odot}$ ,  $R_{\odot}$ , and  $a$ .
- Derive the theoretical effective equilibrium temperature of the planet without an atmosphere  $T$  using  $T_{\odot}$ ,  $R_{\odot}$ ,  $a$ , and  $\alpha$ .



# Assignment (~08<sup>th</sup> Sep)

## □ Planetary Equilibrium Temperature (Cont'd)

Consider a planetary body orbiting the Sun at a distance  $a$  with an albedo  $\alpha$  and radius  $R$ .

The Sun has a luminosity  $L_{\odot}=4\pi R_{\odot}^2\sigma T_{\odot}^4$ , where  $R_{\odot}$  is the radius and  $T_{\odot}$  is the effective surface temperature of the Sun. Solve the following problems:

- In the 1-layer idealized greenhouse model with atmospheric emissivity  $\varepsilon$ , derive the surface temperature  $T_s$  as a function of  $T$  and  $\varepsilon$ , and express the returning longwave radiation re-directed to the surface ( $\varepsilon\sigma T_A^4$ , where  $T_A$  is the effective temperature of the atmosphere).

# Assignment (~08<sup>th</sup> Sep)

## □ Planetary Equilibrium Temperature (Cont'd)

Consider a planetary body orbiting the Sun at a distance  $a$  with an albedo  $\alpha$  and radius  $R$ .

The Sun has a luminosity  $L_{\odot} = 4\pi R_{\odot}^2 \sigma T_{\odot}^4$ , where  $R_{\odot}$  is the radius and  $T_{\odot}$  is the effective surface temperature of the Sun. Solve the following problems:

- Calculate the effective equilibrium temperature for Venus ( $\alpha_V = 0.750$ ,  $a_V = 0.723a_0$ )  $T_V$  and compare its greenhouse effect based on its actual surface temperatures ( $T_{s,V} = \sim 737$  K).

| Parameter   | Value                                                   |
|-------------|---------------------------------------------------------|
| $R_{\odot}$ | $6.96 \times 10^8$ m                                    |
| $T_{\odot}$ | 5780 K                                                  |
| $\sigma$    | $5.67 \times 10^{-8}$ W m <sup>-2</sup> K <sup>-4</sup> |
| $A_0$       | $1.50 \times 10^{11}$ m                                 |

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